

Design Problem

A1

The Problem

Watering plants is an essential task for maintaining a healthy garden or indoor plants, but it can be a challenge for people who frequently travel. Neglecting to water plants while away from home can result in their untimely demise, which can be discouraging for gardeners and plant enthusiasts. To address this issue, I propose to design a solution to address the needs of these clients.

My Client

Joanna Chan is the primary care taker of several indoor herb plants including mint, basil, and chili, which she cares for as a passion and for cooking. However, her family often travels for up to 3 weeks at a time and it is often hard and inconvenient to find ask others to water the plants for her. A lack of water for the plant is harmful to the plant's well being since these plants need great amounts of water to survive.

In order to come up with a solution, research will have to be done to gather more information on the dimensions of the plant and pot



Fig 1: Photo of client



Fig 2: Photo of plants

Primary research

Dimensions of the plants and area



Fig 3: Location of plants



Fig 4: Front view of plants



Fig 5: Side view of plants



Fig 6: Top view of plants



Fig 7: High angle shot of plants

Top diameter (cm)	14
Base diameter (cm)	12
Height (cm)	12
Material	Terracota

Fig 8: Dimensions of pots

Key quotes from client interview

"I spray and water the plant daily"



"I need something that I can move without much difficulty"

"A device that sprays the plant daily with water for me"

"I travel at most 3 weeks at a time"

"I only harvest small quantities at a time"

"Occasional fertilizers and harvesting is required"

"I travel every few months, at least 4 times a year"

"The soil needs to be rich with nutrients with a good drainage system and the plant needs to be exposed to lots of sunlight"

Secondary research

According to fiskars.com's basil guide, chillipeppermadness.com's chill guide, and thespruce.com's guide for mint, below are ideal conditions for growing all 3 plants together in the same place

A1

Ideal conditions	Consequences if need is unmet
At least 6 hours of direct sunlight	Weaker leaves, loss of aroma
Regular watering, water when top of the soil feels dry	Overwatering causes a root rot, underwatering causes wilted leaves
Well draining soil with rich organic matter	Root rot, stunted growth and fewer leaves
Moderate temperatures 15-30°C	Increases stress on the plant that likely leads to death
Moderate to high humidity (around 50-60%)	Low humidity causes brown, crispy leaves and loss of aroma. High humidity creates breeding ground for pests
Fertilizer depending on plant's growth stage	Reduced growth and fewer leaves
Away from pests	Weakened plant and infestation

Market research

According to Gardenpals.com, over 66% of Americans (approx 200 million) own at least one house plant, with around 73% American households traveling at least once per year. These statistics indicate that this is an existing problem for many indoor plant owners, with people that both travel and own indoor plants being the primary target audience. In addition, the global market for indoor plants projected to grow in the upcoming years, suggesting that this will be a growing problem for the upcoming years.

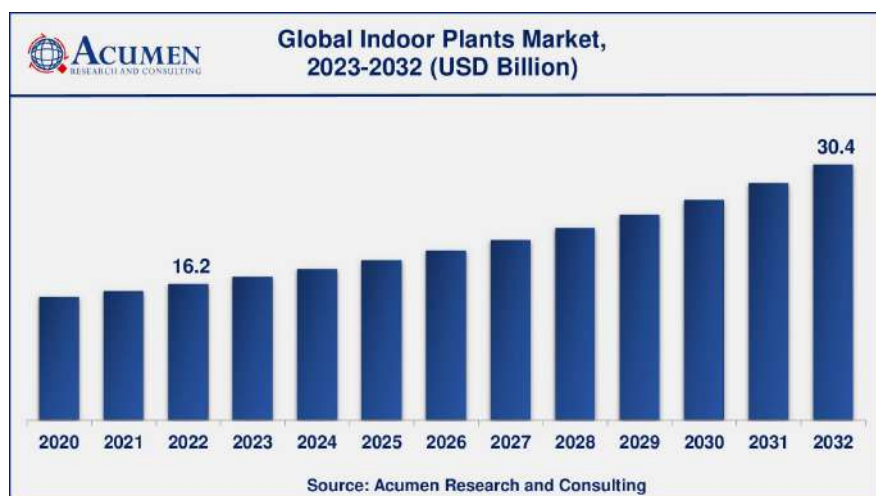


Fig 9: Global market for indoor plants

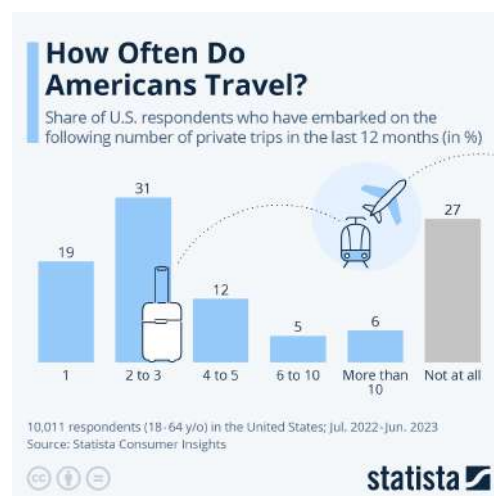


Fig 10: Bar graph indicating how often Americans travel

Product analysis

Below are a list of current products on the market with the advantages and disadvantages evaluated



Fig 11: \$6 Automatic Irrigation Watering System

Advantages

- + Adjustable tap to control the amount of water
- + Easy to use and understand
- + Water lasts for a long time
- + Water can be easily refilled and bottles can easily be swapped
- + Water will not spill out of the pot
- + Easily refillable water

Disadvantages

- One tap could cause a waterlogged spot
- One tap means an uneven distribution of water
- Drip system, hence continuous uncontrolled watering
- Watering system may get in the way of the plant
- Bottle at the risk of tipping over
- May interfere with plans growth



Fig 12: \$66 DIY Automatic Drip Irrigation Kit

Advantages

- + Adjustable system to suit the user's needs
- + System does not interfere with plant growth
- + Easily refillable water
- + Simple user interface
- + Water will not spill out of the pot
- + Does not interfere with plant growth

Disadvantages

- One tap could cause a waterlogged spot
- One tap means an uneven distribution of water
- Exposed water may create a spot for insects to settle
- Battery may die any time which is unreliable
- Exposed water will evaporate



Fig 13: \$75 Hozelock Easy Drip Universal Kit

Advantages

- + Adjustable system to suit user's needs
- + Easy to install
- + Good distribution of water
- + Small and compact system that does not interfere with plant growth
- + Adjustable water capacity
- + Easily refillable water

Disadvantages

- Water strength needs to be monitored
- Problems in tube may be inconvenient to solve
- Limited control over water distribution
- Chance that water may spray out of the pot
- Dimensions may not match with pot

Design Brief

A2

The goal of this project is to design and manufacture a **fully functional high fidelity, 1:1 scale, one-off prototype** of an **automatic plant watering system** that costs **below \$70 SGD**. I will take an **incremental approach** to create a suitable adaption for my product with the target audience being **people who travel away from home frequently** and are **unable to water their plants**. The product should be a **waterproof automatic system** that can be **adjusted** to the **user's needs** to avoid overwatering and underwatering. The product should have the ability to water **multiple plants** at **multiple heights** at the **same time** that are positioned in a **38x28.5x15 cm space** and must be **simple** to use to avoid user error. It must be able to **evenly distribute** water across soil for different plants without any water spilling out of the pot to avoid further problems. and must be able to last for **at least 3 weeks at a time** without refilling water to ensure the plants stay alive. The system **must not interfere** with plant growth by blocking light or space. In addition, water should be easily **refillable** and the product should be waterproof. The product must be ready before **February 2024**.

Design Specifications

A3

Below are design specifications, justification and citation. Specifications are presented in **order of priority**.

Specification	Justification	Evidence
1. Function		
1.1 Product must be an automatic watering system	"A device that sprays the plant daily with water for me." This will solve the client's problem of being unable to water plants when out of the country	Key quotes from client interview
1.2 Product should have the ability to water 3 plants on different levels	"Several indoor herb plants including mint, basil, and chili." The client has 3 plants at multiple heights, hence, the product should be able to water plants at different heights	My client, Fig 2: Photo of plants
1.3 Product should be timed and adjustable to the user's needs	"I spray the plant daily", "an automatic system that should be able to be adjusted to the user's needs to avoid overwatering and underwatering." An adjustable system is a system that can match the user's needs	Key quotes from client interview, design brief

Design Specifications

A3

Specification	Justification	Evidence
1.4 Product should evenly distribute water on soil	Having the water evenly distributed will ensure that no spot is getting waterlogged and ensure that the none of the soil dries out	"10 Golden Rules for Watering" by Gardena
1.5 Water must be easily refillable	The water must be refillable so that the product can be reusable. This will avoid any inconveniences and is a fundamental function of an auto watering system	"What are self watering planters and how do they work?" by greenery unlimited
1.6 Water should be able to last at least 3 weeks	"I travel at most 3 weeks at a time." The water should be able to last at least 3 weeks on a setting that provides enough water for the plant	Key quotes from client interview
2. Product constraints		
2.1 Product must not hinder plant growth by blocking space	"The system must not interfere with plant growth by blocking light or space." The product must not cause any new problems for the client and should be a solution	"Environmental Factors Affecting Plant Growth" by Oregon State University
2.2 Product must not block the plant's sunlight	The product must not interfere with plant growth. Plants need sunlight to grow and survive	"Lighting for Indoor Plants and Starting Seeds" by University of Minnesota Extension
2.3 Product must not be over 10kg	"I need something that I can move without much difficulty." The product will have to be moved occasionally for cleaning or repositioning and should not be unnecessarily heavy	Key quotes from client interview
3. Aesthetics		
3.1 Product must have a tank that can hold at least 2 litres of water	"I travel at most 3 weeks at a time." A litre of water should be able to provide a sufficient amount of water for the plants for when the client is on holidays	Key quotes from client interview
3.2 Product must not have any exposed circuits	Circuits must be kept enclosed in the product to avoid damaging the circuit or harming the user, especially since water is part of the system	"Physical Electrical Hazards" by Canadian Centre for Occupational Health and Safety
3.3 Water must be covered and should not be exposed	This will prevent water from evaporating or prevent any unwanted insects or objects from entering the tank	"Evaporation in your Aquarium" by Bulk Reef Supply
3.4 User interface must be simple and easy to understand (if there is one)	A simple user interface will allow the user's needs to be met efficiently and will save the user some time providing a good user experience	"14 Benefits of Good User Interface Design" by Alaina Roberts on Digital Doughnut

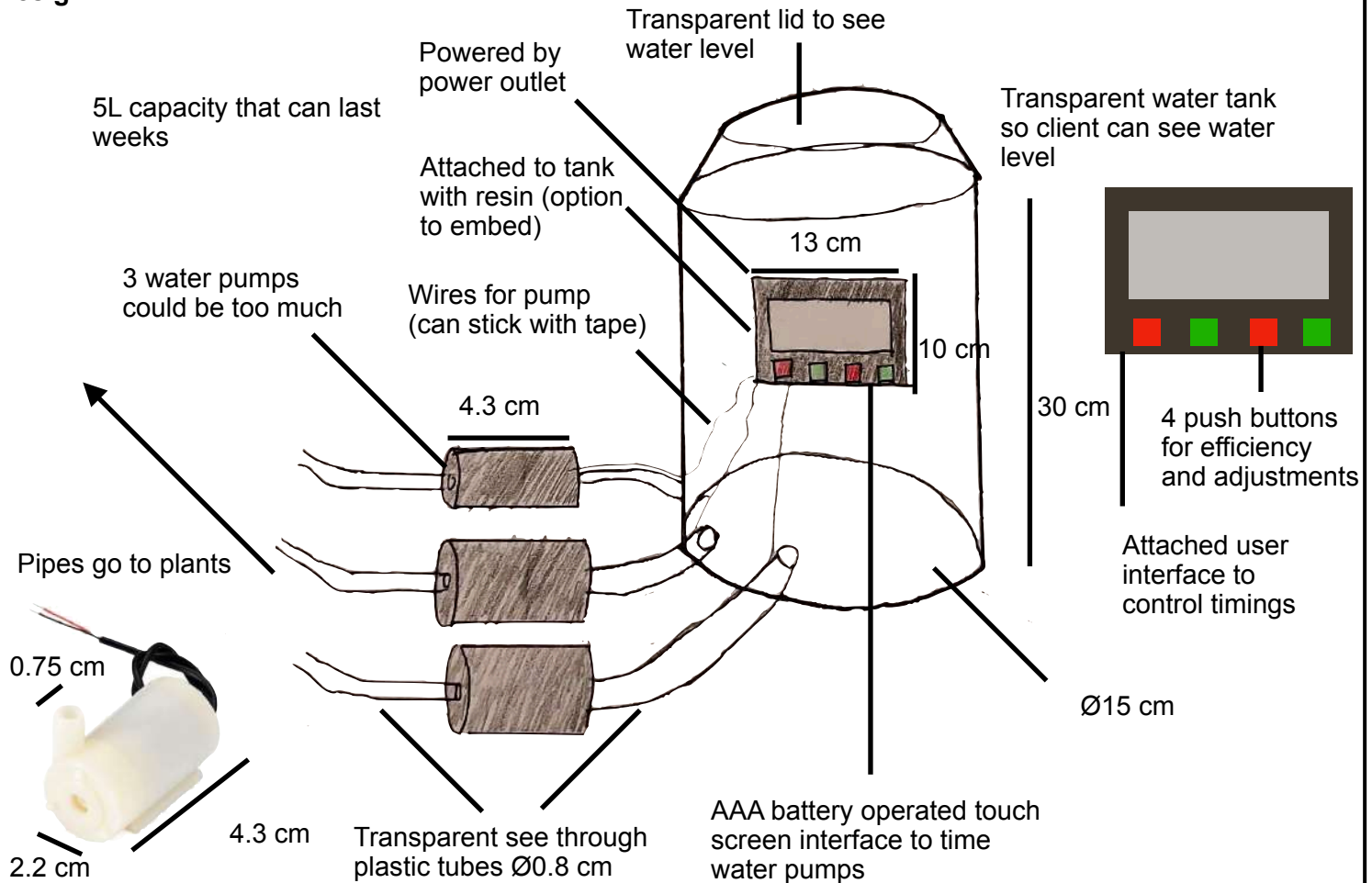
Design Specifications

A3

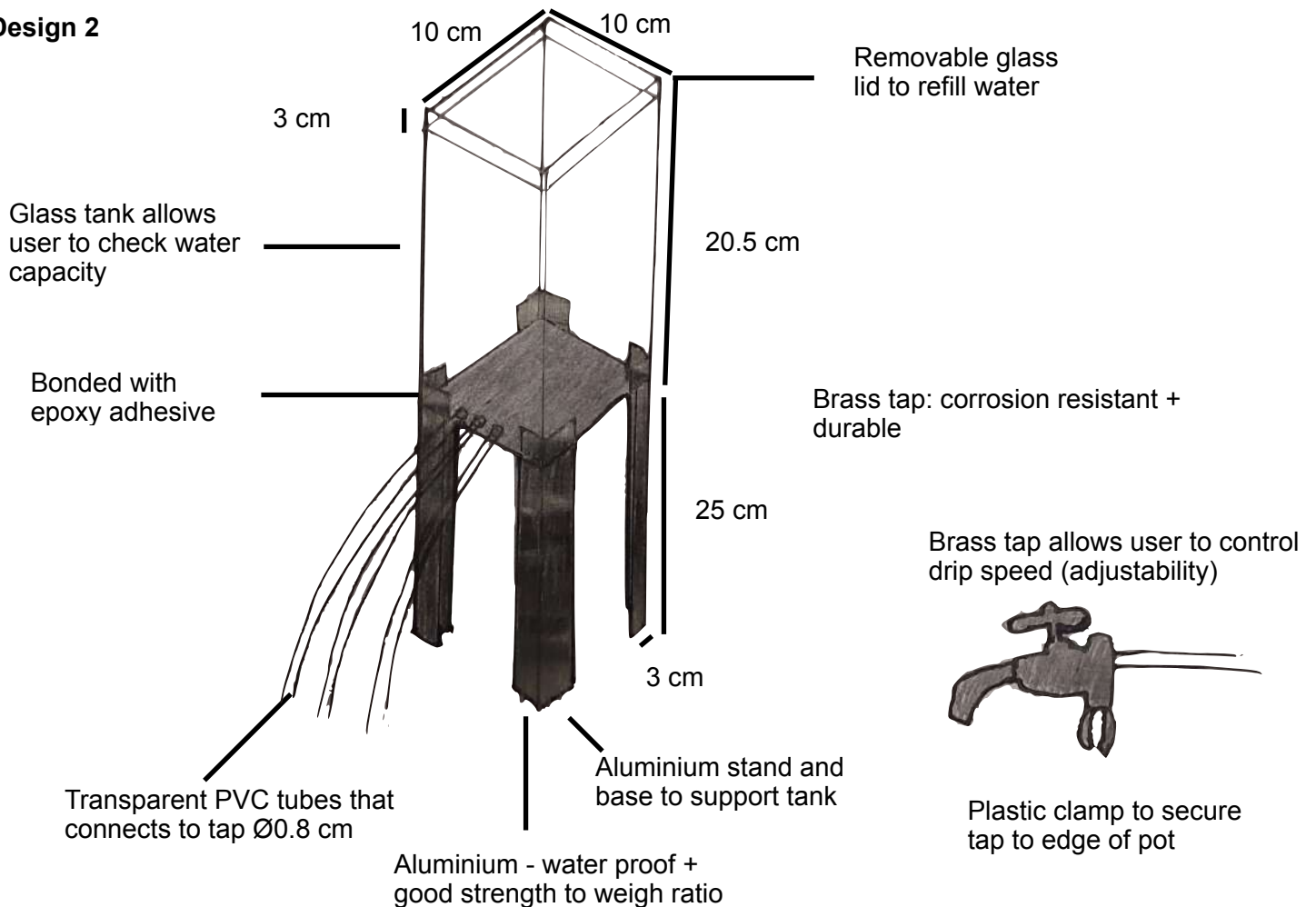
Specification	Justification	Evidence
4. Material selection		
4.1 Material must be waterproof	The product will get wet as it is an auto watering system. The material must be waterproof so that no harm will be done to the product	"Water Damage on Building Materials" by Juvenile
4.2 Material must be able to withstand sunlight	The material must be able to withstand sunlight as the product will be placed next to a window. The product must not be affected by sunlight to maintain its quality	Dimensions of the plant and area, <i>Fig 3: Location of plants</i>
4.3 Material must be strong and durable	The material should be strong and durable to such that the product will withstand external forces and be able to last long so that it provides a better user experience	"Why is Product Durability Important?" by Excelsior
4.4 Material must not be toxic	The product must not harm the plant's well being and hence should not be made of a material that may be hazardous to the plant	"Plant disease" by Britannica
5. Size		
5.1 The product (including tank) must not be longer than 55 cm	The length of the product must not exceed 55 cm as this is the length of the space. The height and width are not strictly constrained but should not exceed 50 cm as it is unnecessary for it to be that large	Dimensions of the plant and area, <i>Fig 7: High angle shot of pots</i>
5.2 The system must provide water to plants in an area of 38 x 28.5 x 27 cm	The rack which the plants are located in is 38 cm wide and 28.5 cm long. The highest pot is 15 cm high and the pot is 12 cm tall. The product must be able to provide water to this zone	Dimensions of the plant and area, <i>Fig 3: Front view of plants, Fig 6: Top view of plants, Fig 8: Dimensions of pots</i>
6. Target audience		
6.1 Product is for people who own indoor plants and frequently travel	"it can be a challenge for people who frequently travel." This product aims to solve problems for people who own plants and travel frequently	The problem
7. Competition		
7.1 Product must be below \$70	A cheaper product that performs its job will be an ideal substitute for other products already in the market. A \$70 product that meets the specifications in functions is a superior alternative to products listed in the product analysis	Product analysis
7.2 Product must be easy to understand and use	A product that is easy to use will reduce use error and misuse of the product. It will reduce inefficiency as the client knows how the product works and will help the client save time	"The benefits of simple product design in a market" by Michael Hemsworth on trendreports.com
8. Quantity		
8.1 The product is a 1:1 scale fully functional, one-off prototype	This is a one-off product due to its customisation, complexity, time and resources	"Manufacturing & Methods of Production" by Designkmg

Concept Sketches

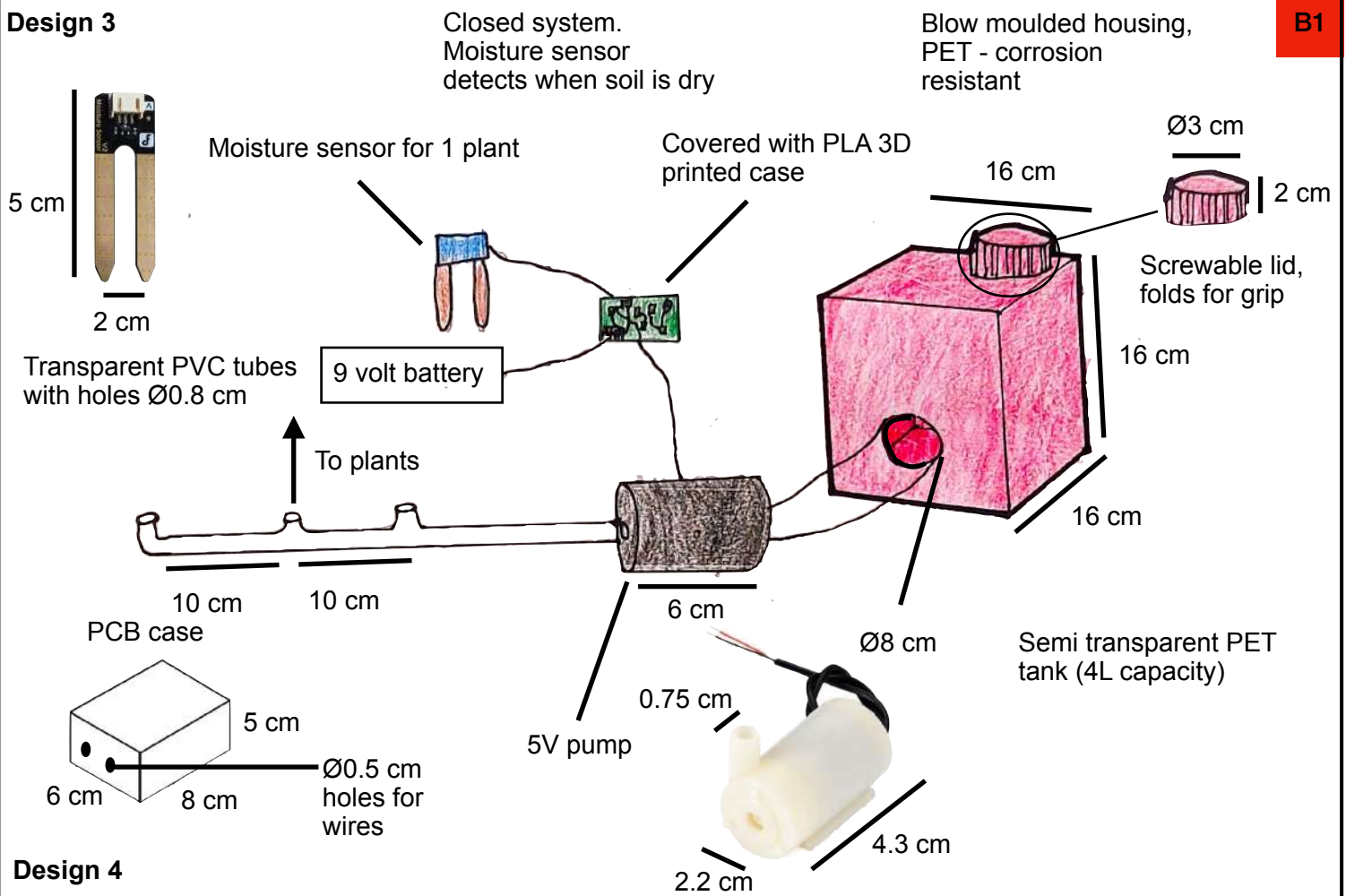
Design 1



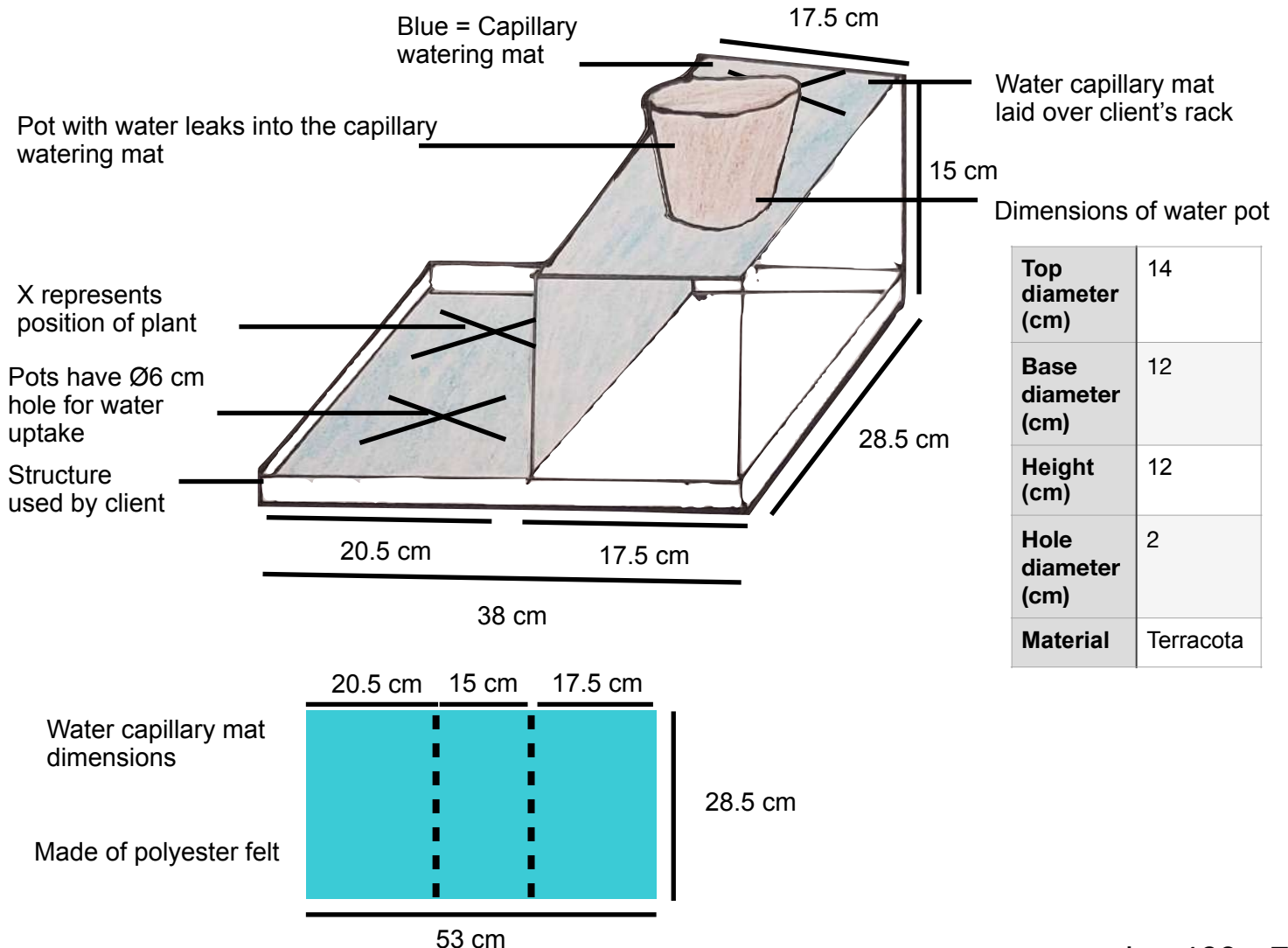
Design 2



Design 3

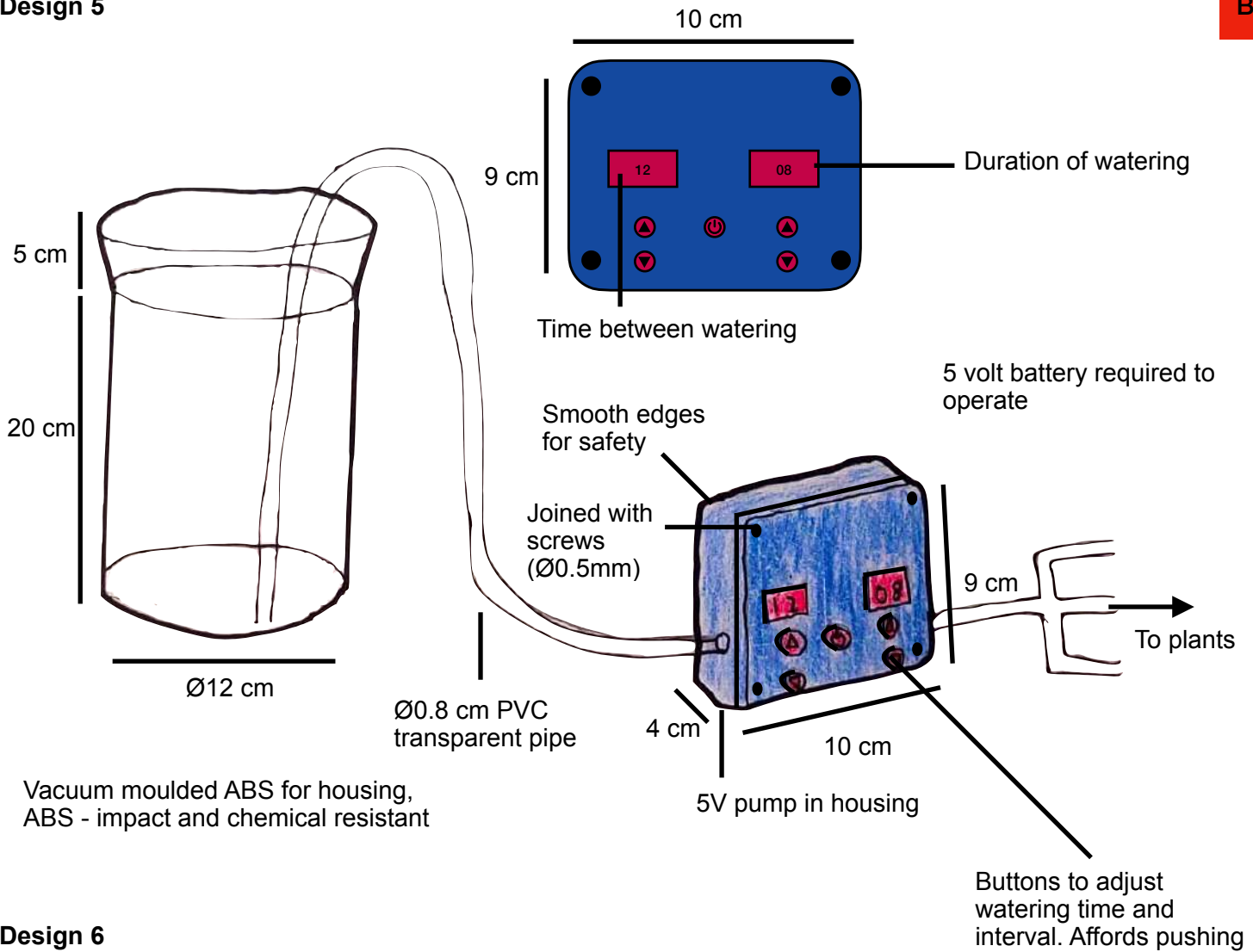


Design 4

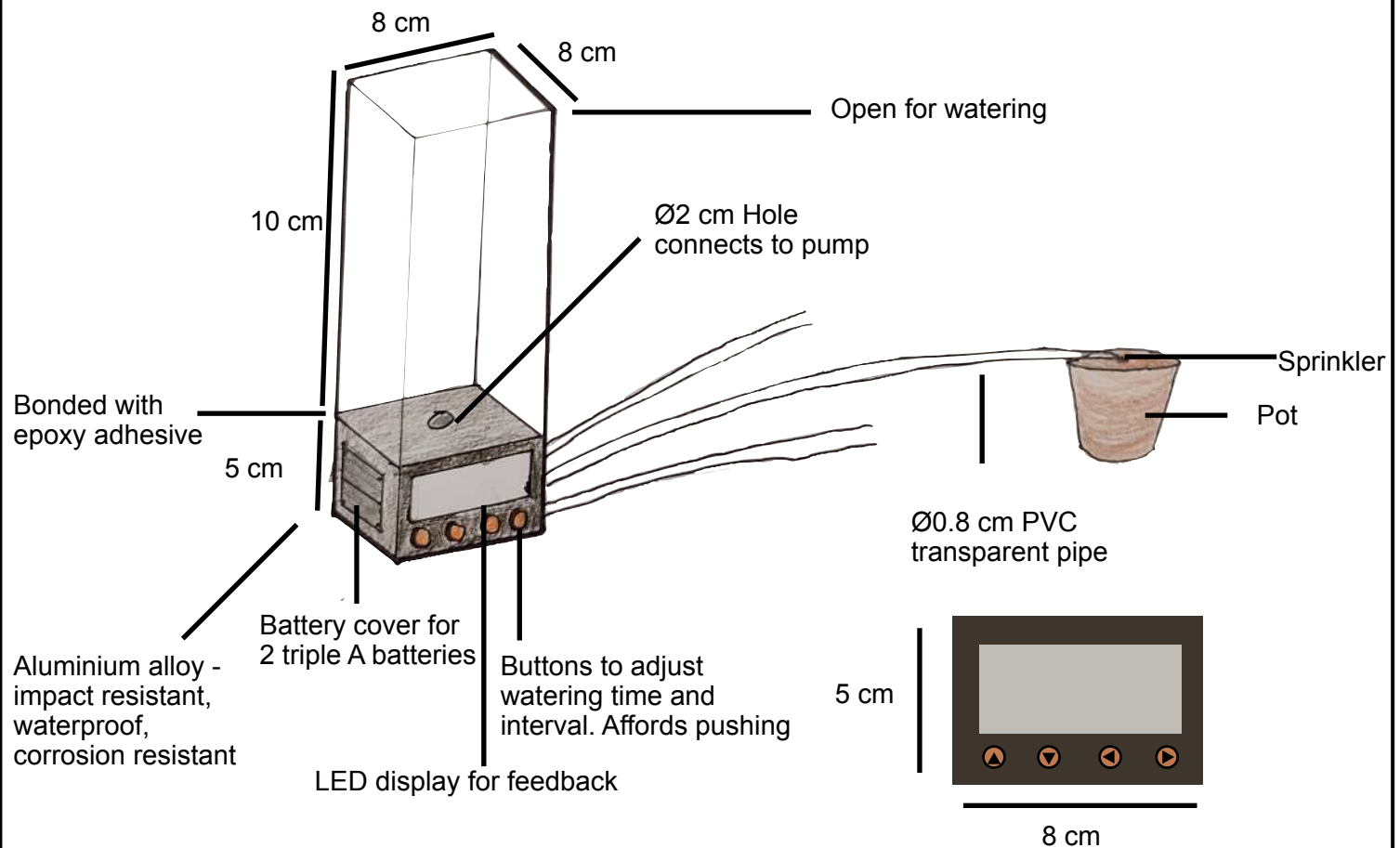


Design 5

B1



Design 6



Review of Conceptual Designs

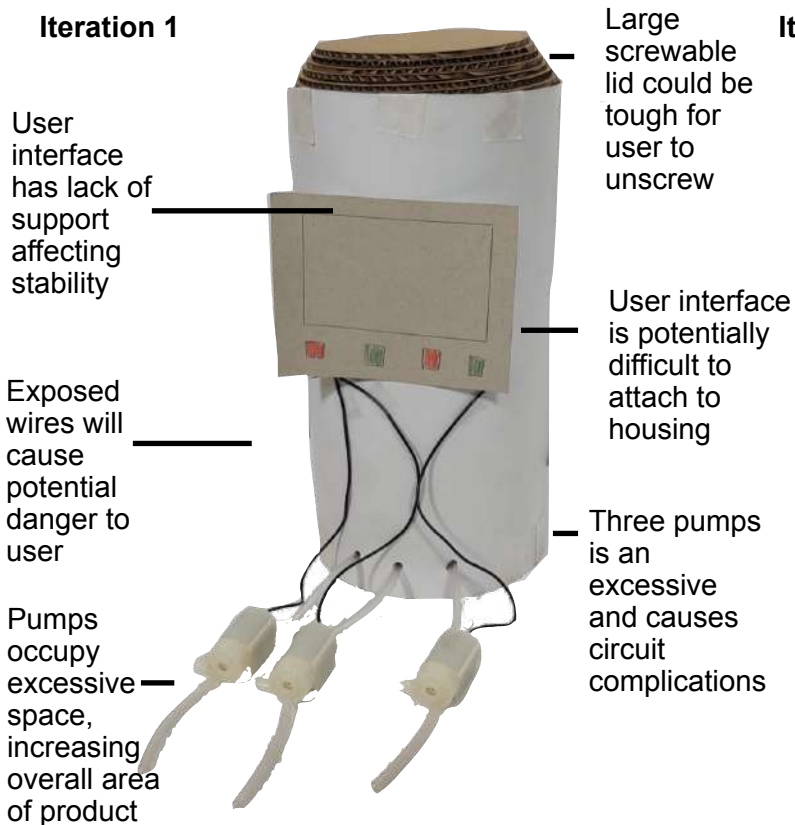
B1

Designs highlighted in green will be the designs that are further developed

Design	Strengths	Potential issues
1	- Size: Stable water tank which won't topple - Function: Easily controllable pumps	- Constraints: Exposed pumps will have to be handled with care - Aesthetics: Body is curved but user interface is not which will be problematic for attachment
2	- Function: No electricity used - Function: Taps easily adjustable to user's preference	- Size: Risk of design toppling over - Function: Constant drip may lead to overwatering - Material selection: Lid may fall over and break if not handled carefully
3	- Function: Automatic, feedback from plant (moisture) is obtained for when to water - Constraints: Easy to assemble	- Function: Watering hole may be too small and hard to fill - Function: Water not distributed evenly due to pipe structure - Constraints: Exposed components will have to be handled with care
4	- Function: No electricity used - Function: Waters plant according to moisture level	- Function: Could be troublesome to maintain and repair if there is failure - Aesthetics: Hole will have to be put into pots - Function: Water will evaporate from the mat
5	- Size: Compact design and user interface - Function: Water tank interchangeable with other containers	- Function: Exposed water which will lead to evaporation - Size: User interface will have to be handled with care due to lightness
6	- Function: Easily accessible user interface and batteries - Size: Stable water tank which won't topple - Function: Even distribution of water	- Function: Exposed water which will lead to evaporation - Constraints: Difficult to assemble

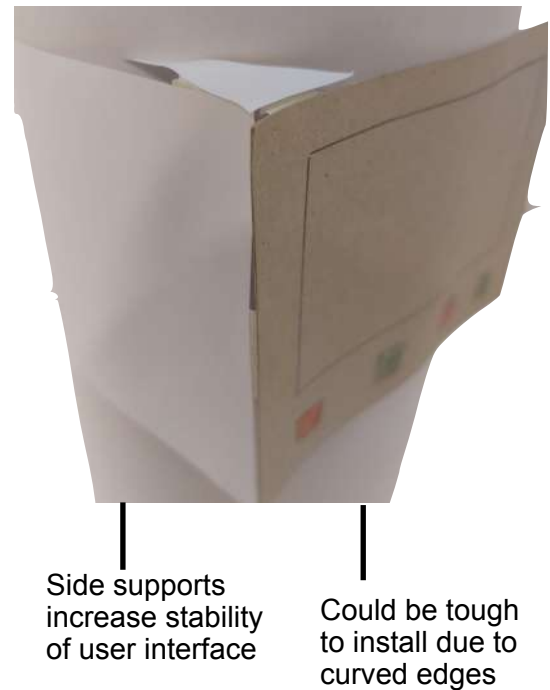
Development of idea 1

Iteration 1



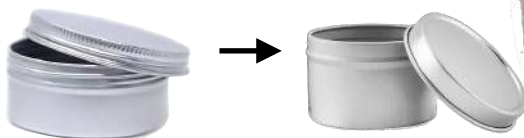
Iteration 2

Increased support for user interface



Iteration 3

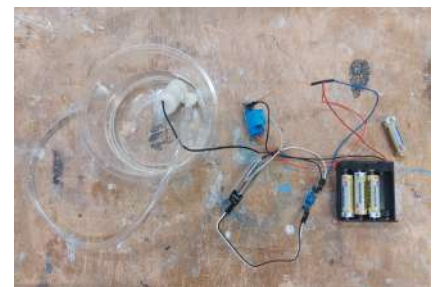
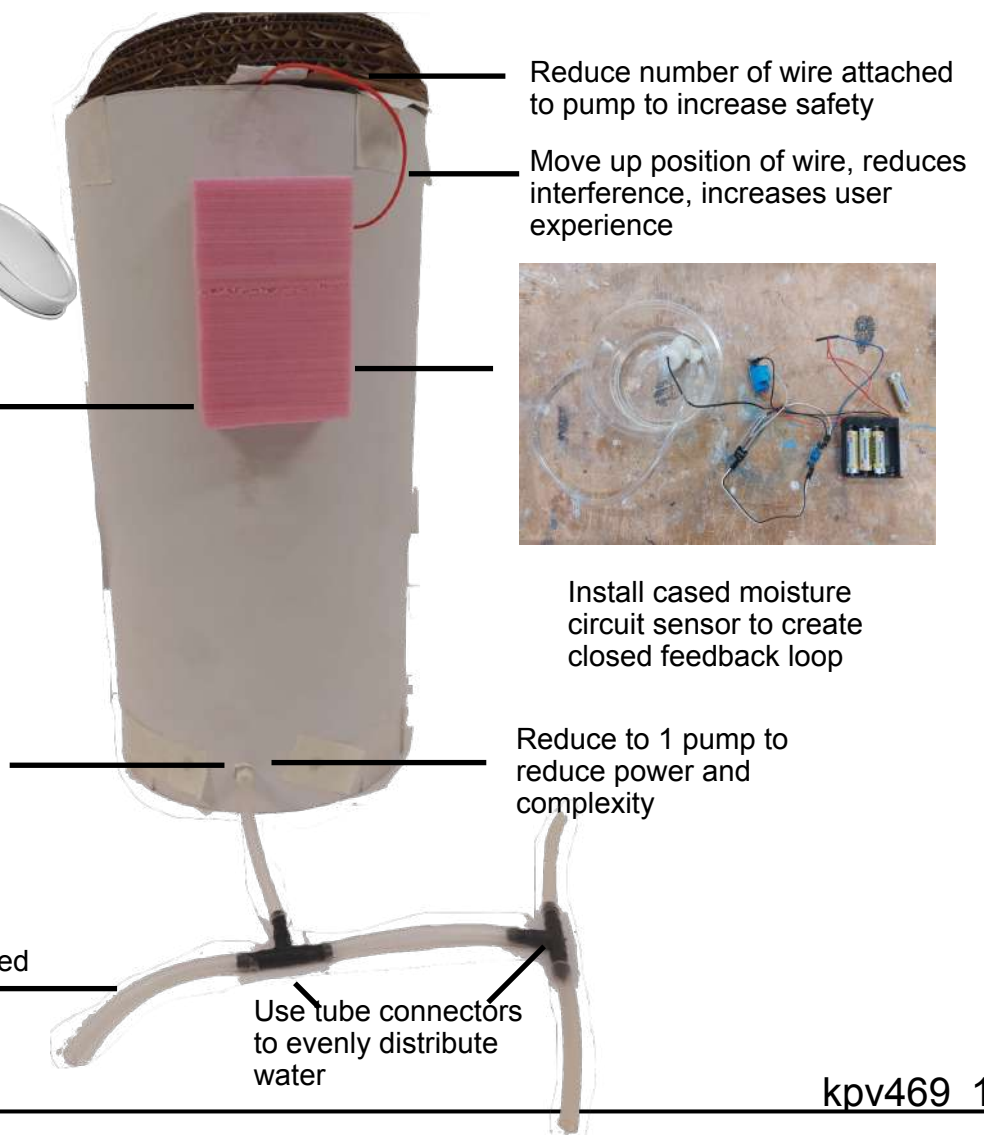
Change screwable lid to seal lid to increase simplicity



Simple compact PCB case with screws for unlocking

Move pump inside tank total area occupied

Increase length of tubes for increased flexibility and user experience



Development of design 2

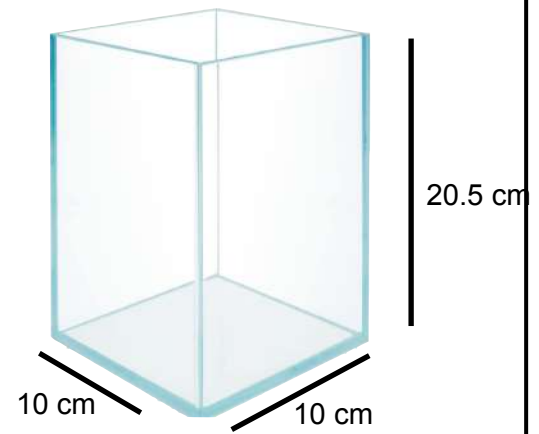
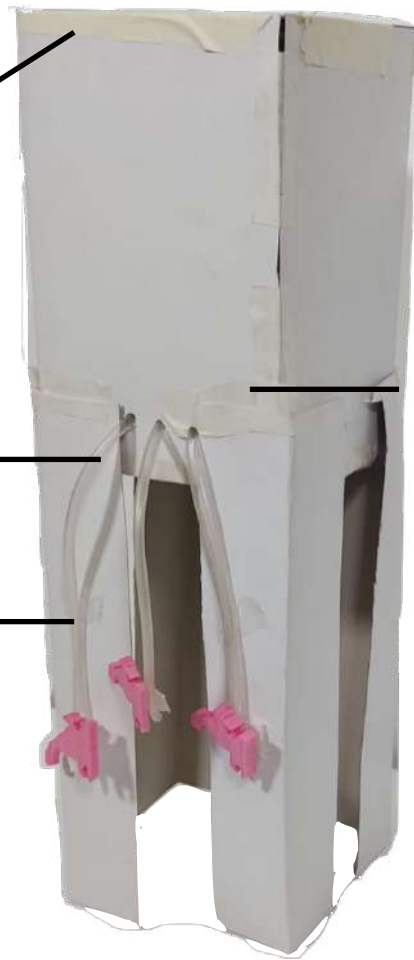
Iteration 1

Detachable lid will fall off easily

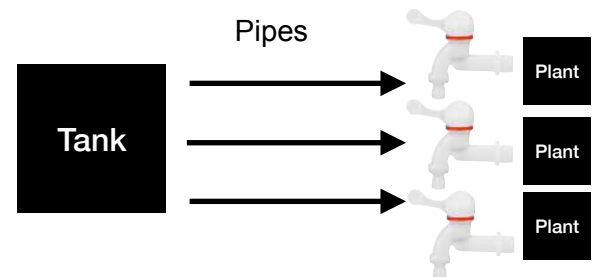


Potentially unstable base, which might affect user safety

3 pipes are redundant, reduced user experience



Glass tank (without lid)



Iteration 2

Change to hinged lid rather than removable lid

Surface model (model not including pipe)



Removable lid

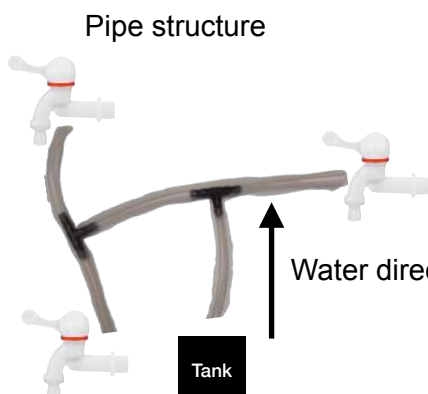
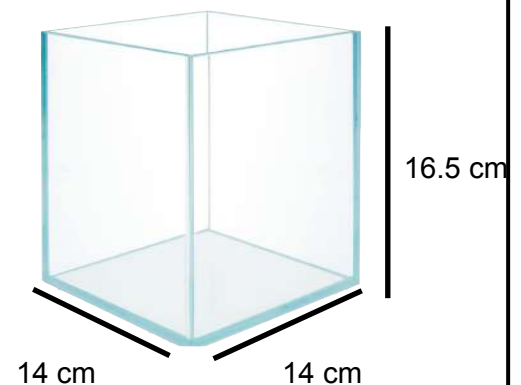
+ Replaceable
- Losable

Hinged lid

+ Always attached to product (better efficiency)
- Harder to clean

34.5 cm

Reduced height of tank, increased width of tank



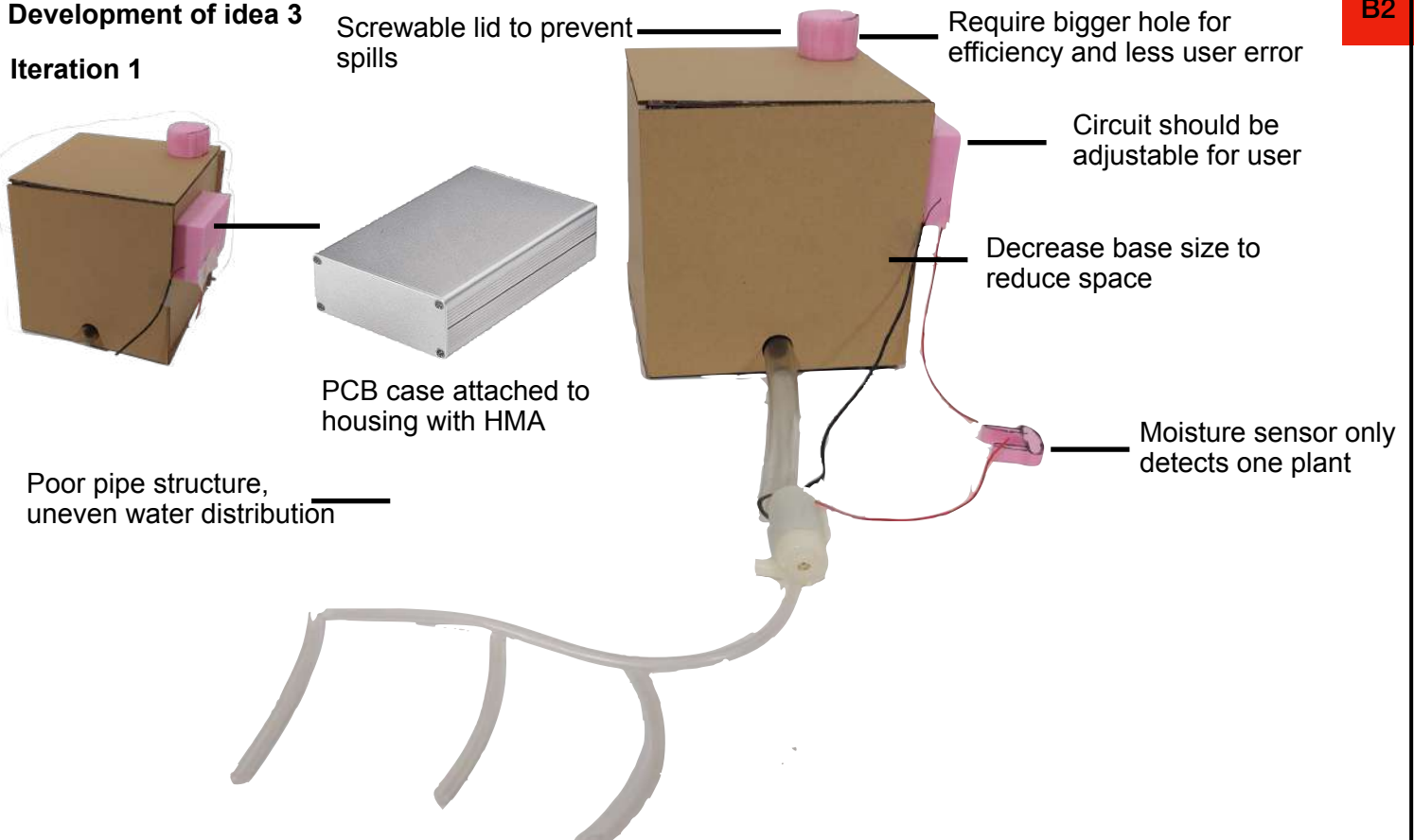
4 cm

Increased width of legs, decreased height

Increased stability, reduced risk of collapse

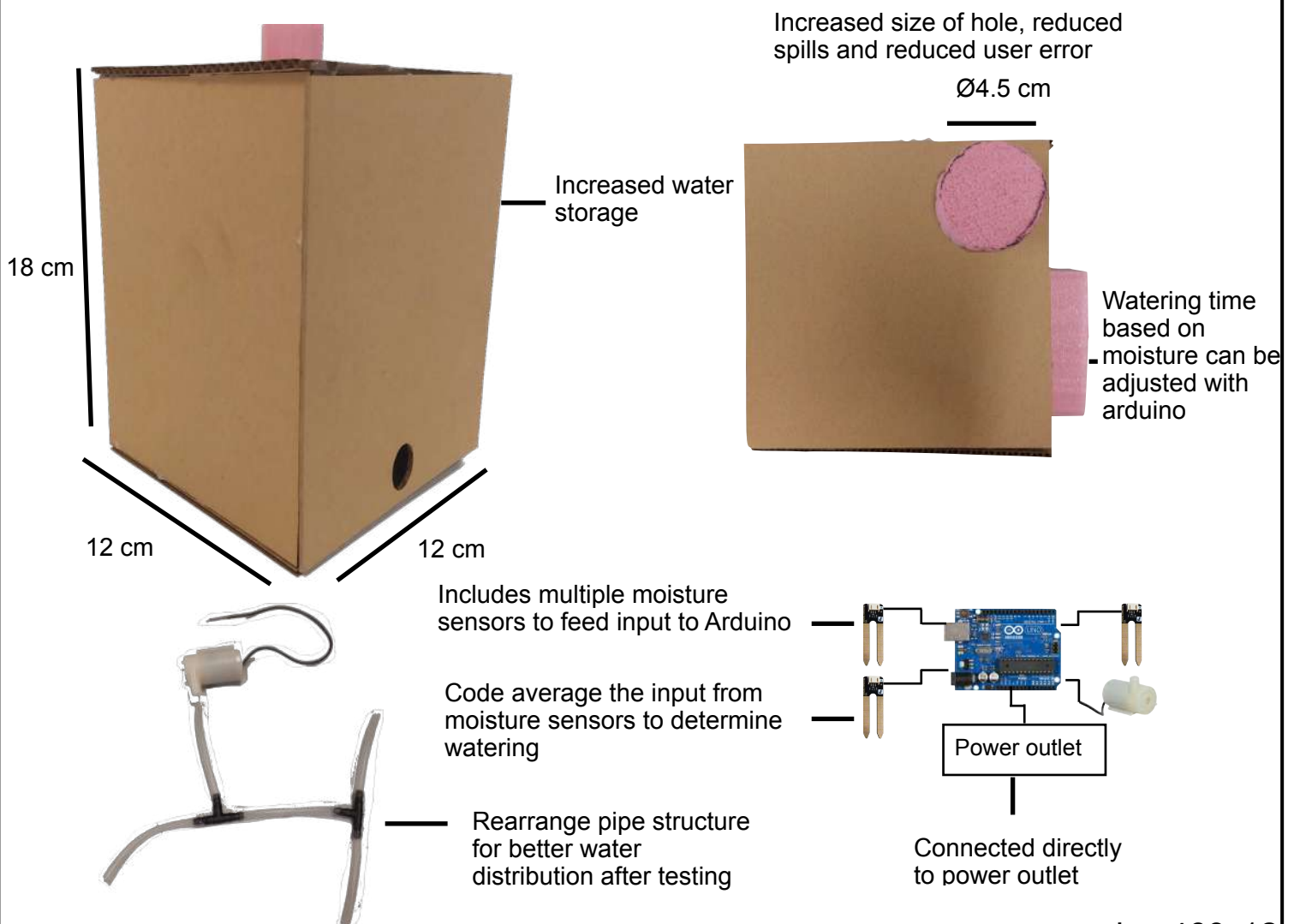
Development of idea 3

Iteration 1



Iteration 2

Change shape of body to rectangular prism



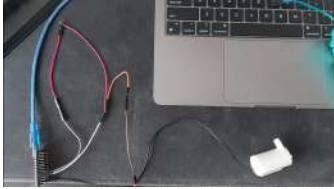
Review against Design Specification

B2

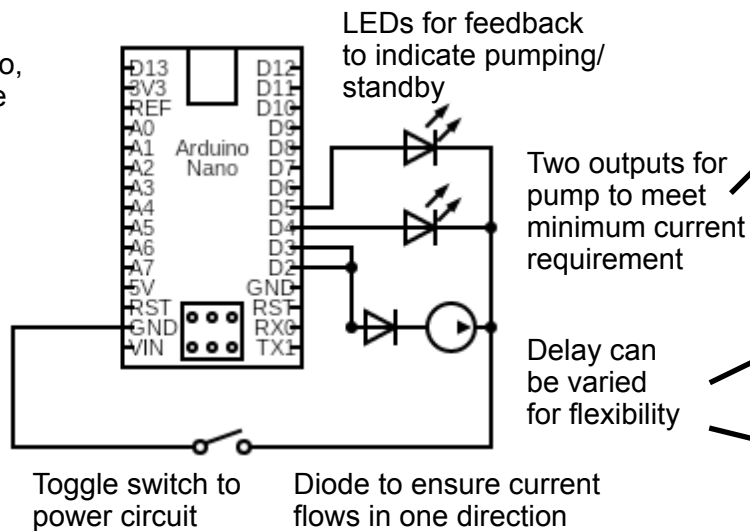
Design specification	Design 1	Design 2	Design 3
1.1 Product must be an automatic watering system			
1.2 Product should have the ability to water 3 plants on different levels			
1.3 Product should be timed and adjustable to the user's needs			
1.4 Product should evenly distribute water on soil			
1.5 Water must be easily refillable			
1.6 Water should be able to last at least 3 weeks			
2.1 Product must not hinder plant growth by blocking space			
2.2 Product must not block the plant's sunlight			
2.3 Product must not be over 10kg			
3.1 Product must have a tank that can hold at least 2 litres of water			
3.2 Product must not have any exposed circuits			
3.3 Water must be covered and should not exposed			
3.4 User interface must be simple and easy to understand (if there is one)			
4.1 Material must be waterproof			
4.2 Material must be able to withstand sunlight			
4.3 Material must be strong and durable			
4.4 Material must not be toxic			
5.1 The product (including tank) must not be longer than 55 cm			
5.2 The system must provide water to plants in an area of 38 x 28.5 x 27 cm			
6.1 Product is for people who own indoor plants and frequently travel			
7.1 Product must be below \$70			
7.2 Product must be easy to understand and use			
8.1 The product is a 1:1 scale fully functional, one-off prototype			

Development 1

Changed circuit to Arduino Nano, moisture sensor was insensitive



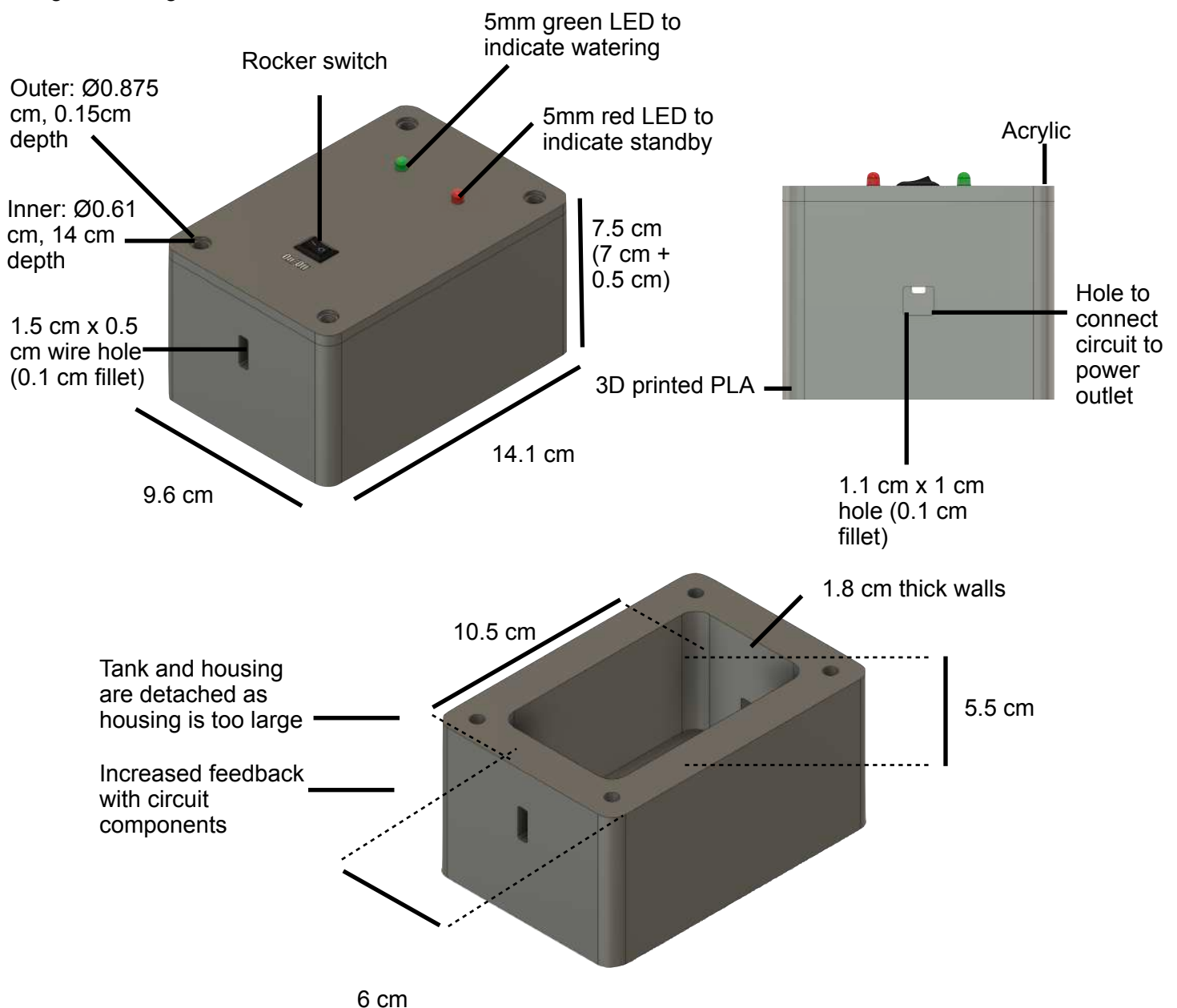
Testing of circuit



```
1 void setup() {
2   pinMode(2, OUTPUT);
3   pinMode(3, OUTPUT);
4   pinMode(4, OUTPUT);
5   pinMode(5, OUTPUT);
6 }
7 void loop() {
8   digitalWrite(2, HIGH);
9   digitalWrite(3, HIGH);
10  digitalWrite(4, LOW);
11  digitalWrite(5, HIGH);
12  delay(5000);
13  digitalWrite(2, LOW);
14  digitalWrite(3, LOW);
15  digitalWrite(4, HIGH);
16  digitalWrite(5, LOW);
17  delay(1000);
18 }
```

Development 2

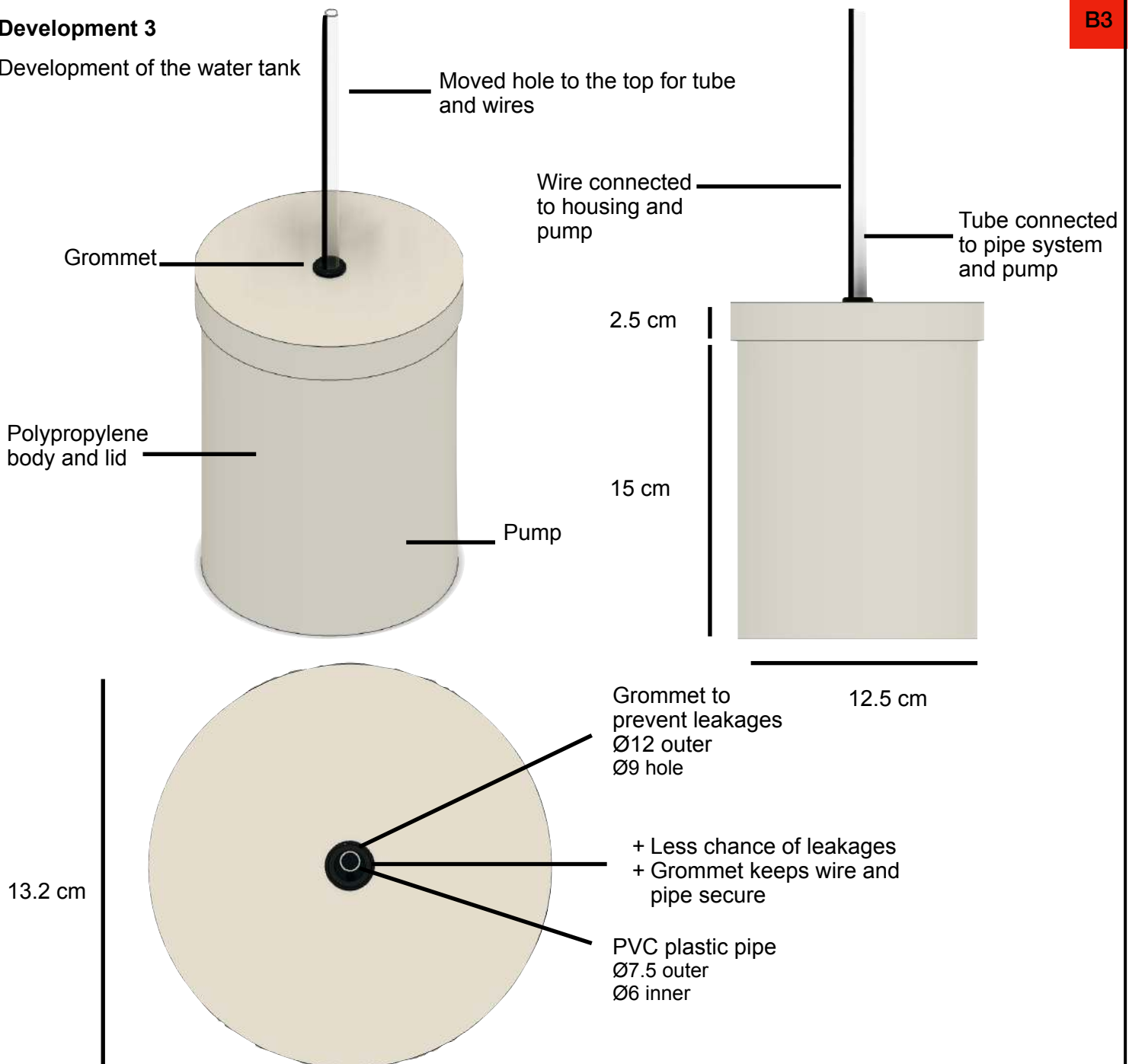
Separated housing from tank and enlarged housing



Development 3

Development of the water tank

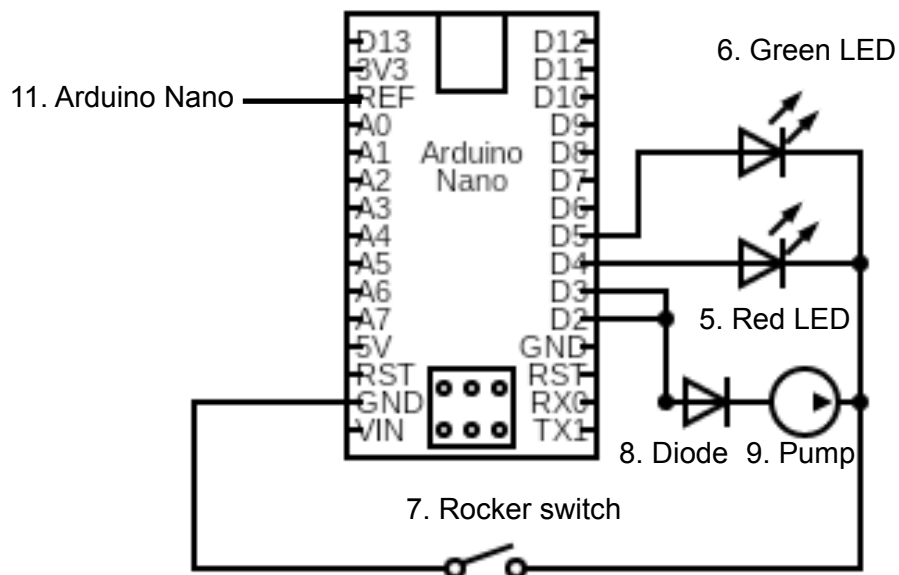
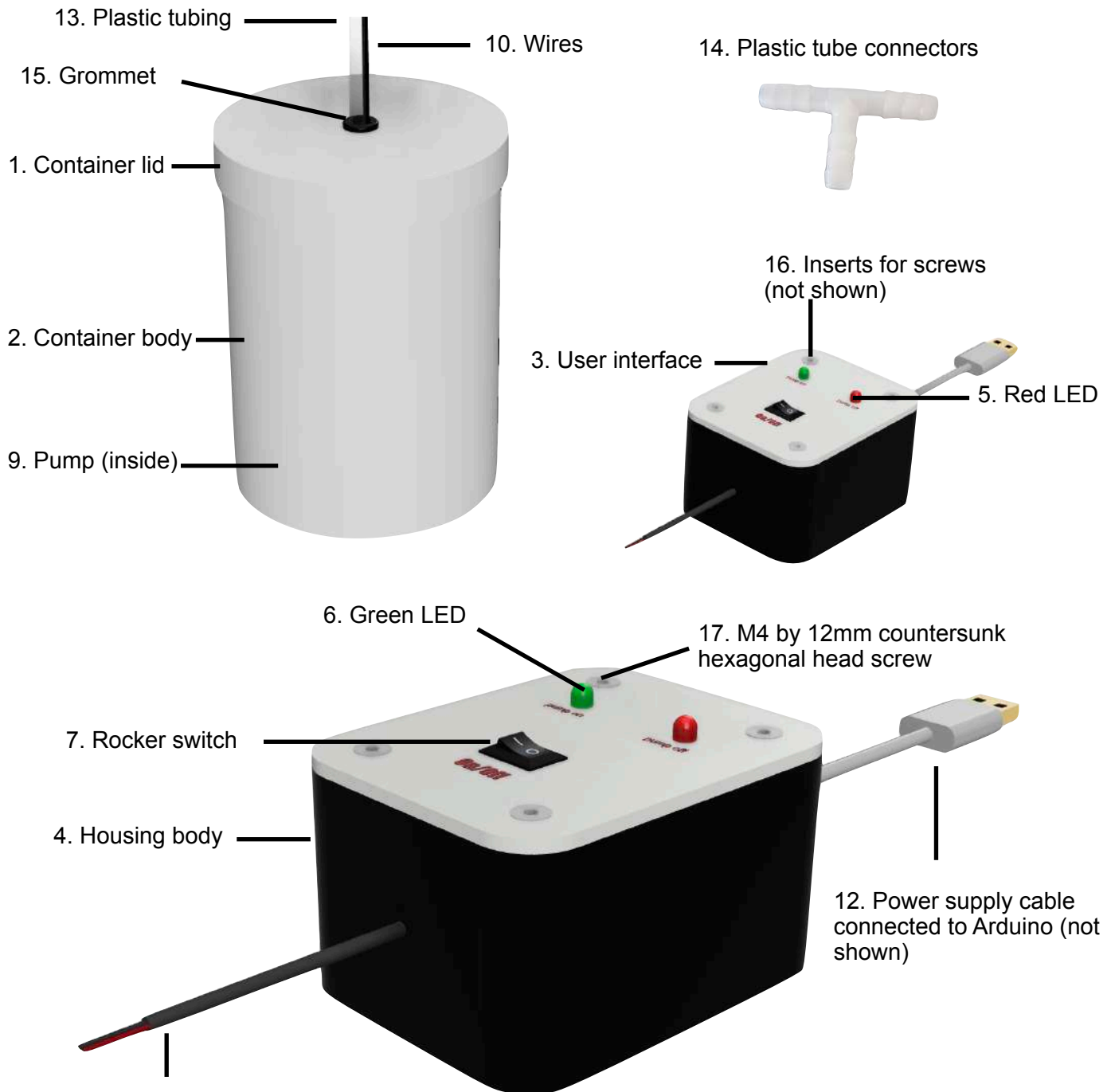
B3



The design that will be moving forward will be design 1. Although all 3 designs meet a similar number of specifications, the concept of this design combined with the housing is the most straightforward in **functionality** and usability as it **doesn't involve complicated processes** involving controlling individual plants yet still achieves the function. This product is **unique** compared to those on the market as it requires **minimal interference** from the user whilst being able to water 3 plants at the same time. The Arduino circuit makes it such that it allows the user to change the timings if needed. Furthermore, the user interface makes it such that it is **straightforward** for the user to understand, with a power switch and two 5mm LEDs to indicate the **status** of the system. The system will be powered via a power outlet as opposed to a battery which is crucial as it prevents the system from stopping whilst the user is on holiday where they are unable to change it. There are wires coming out of the front of the housing connecting the Arduino to the pump via a shared hole with the tubing in the roof of the water tank which the user will have to be careful with. Having the hole at the top of the water tank rather than at the side aids **functionality** as water cannot spill out of the tank. Having a screwable lid also makes it **efficient** for the user to refill water and prevents water from evaporating out of the tank. The housing for the circuit will also be small and **compact** with the proposed tank for the water taking up **adequate space**. The main body is made out of 3D printed **PLA** which is **waterproof** and has good **impact resistance**, whereas the user interface is to be made of acrylic, which is **lightweight** and **waterproof**.

Justification of Materials and Components

C1



Justification of Materials

Part name: Housing	Material selection: PLA (Polylactic Acid)	Part number: 3
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Justification of physical materials: PLA is suitable for the process I will use to create the housing for the arduino nano (3D printing). Its affordability and ease of 3D printing make PLA a practical choice for crafting customized housing. PLA is also lightweight meaning it can be moved easily yet it is durable and strong enough to withstand external forces to protect internal components. PLA also exhibits good thermal stability, which is crucial for maintaining the integrity of the circuitry and preventing any potential heat-related issues.

Environmental considerations: PLA is known for its eco-friendly properties, as it is derived from renewable resources such as corn starch or sugarcane. This assures that the product is sustainable as the material used for the housing is environmentally responsible.

Part name: User interface	Material selection: Acrylic	Part number: 4
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Justification of physical materials: Acrylic has been selected as the ideal material for the user interface due to its versatility and ability to provide a clean and sleek appearance. Acrylic comes in many colours which will allow for flexibility in choice and the ability to choose white to contrast the black PLA. Acrylic's durability ensures that the user interface maintains its pristine white appearance over time, resisting scratches and wear. Moreover, the ease of laser cutting through a sheet of Acrylic allows for an easy accurately made user interface.

Environmental considerations: Although Acrylic is derived from petroleum, Acrylic is durable and is a type of thermoplastic meaning it can be recycled after use.

Part name: Water tank	Material selection: Polypropylene	Part number: 1, 2
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Justification of physical materials: Polypropylene has strong chemical resistance which minimizes the risk of corrosion or degradation. Polypropylene boasts high durability and impact resistance, safeguarding the water tank against external forces or accidental impacts. It is also lightweight enabling easy handling and installation, enhancing the overall user experience.

Environmental considerations: Polypropylene is a recyclable thermoplastic, allowing for the option for responsible disposal or reuse. Polypropylene also produces minimal toxic and carbon emissions.

Justification of Manufacturing Processes

Part name: Housing	Process name: 3D printing	Part number: 4
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Justification of manufacturing process: 3D printing, specifically Fused Deposition Modelling (FDM) and Stereolithography (SLA), was appropriate due to access to a 3D printer and its compatibility with PLA. 3D printing was also suitable for precisely producing a body from a given CAD (fusion 360) design. Many parts of the body required precise production such as the 3mm wire hole and screw holes. Furthermore, the shape of the body was easy to 3D print given its compactness and cuboid like shape.

Part name: User interface	Process name: Laser cutting	Part number: 3
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Justification of manufacturing process: Laser cutting was suitable for the user interface as it required precise shapes for the components to fit in. This was easily done online through 2D drawings. Furthermore, laser cutting works well with acrylic as it doesn't suffer from thermal expansion which is key in maintaining a face.

Part name: LED, rocker switch, pump, diode	Process name: Soldering, Glueing, Heat Shrinking	Part number: 5, 6, 7, 8, 9, 10, 11
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Justification of manufacturing process: Soldering is appropriate given the components at hand. Soldering ensures there are strong connections between components. Glueing was also used to join connections that could not be joined with solder such as connections between an LED and a female to female wire, which has been used due to its efficiency with the Arduino Nano. Heat shrinks have also been used to prevent the circuit from short circuiting and has been done using a lighter due to the limited amount of wire being needed to be heat shrunk.

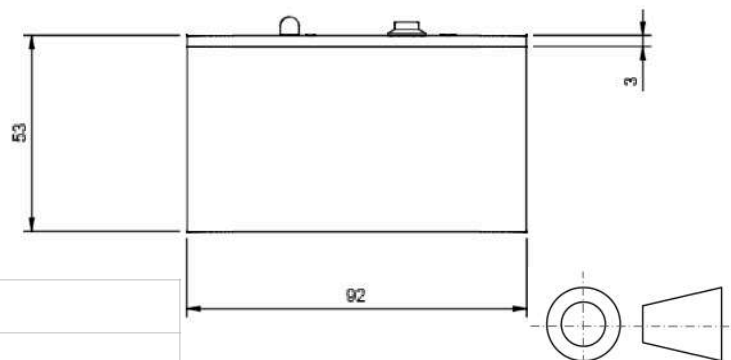
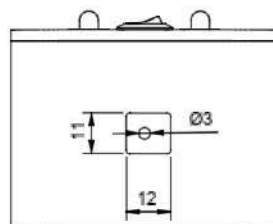
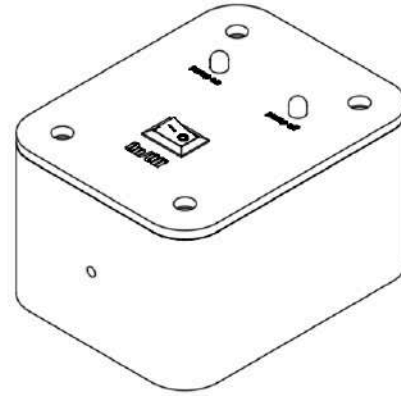
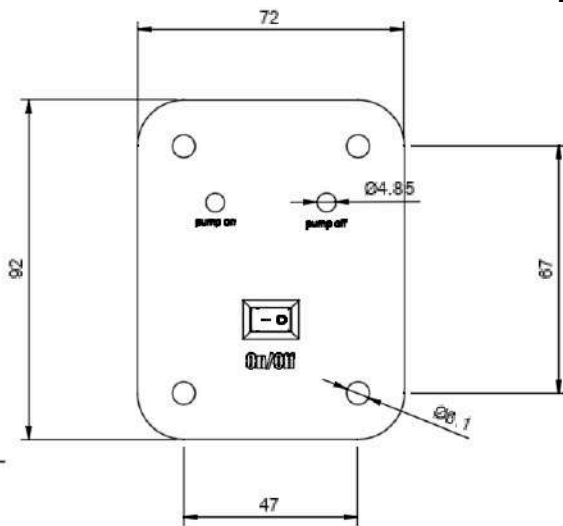
Justification of Standard Components

Arduino nano	The Arduino Nano has been selected due to the need for creating a timed loop with a period of watering and a period of pause. This need was simply met through coding. The Arduino Nano also outputted the appropriate voltage and current of 5V and 90mA which was enough for the components to run.
LED	LEDs have been selected to indicate whether the pump is on or off. They last 50,000 hours and require a voltage of 2V and a current of 20mA.
Diode	The diode has been used to ensure that current is only running in one direction. The diode was implemented after discovering an issue with the circuit where the red LED would light up despite the rocker switch being open.
Rocker switch	The rocker switch has been used to allow the user to power the product by breaking current flow when open.
Pump	The pump has been used to control water flow from the tank to the pipe and to the plants. The pump requires 3V and 130mA to operate which is why it has been put in parallel from two Arduino outputs (2x90mA) to meet the required current.
Water tank	The water tank has been selected as it is required as a water supply for the plants. This specific tank has been selected since it has a good capacity and is semi-transparent which will allow the user to see the water level as well as not absorb too much sunlight

Bill of Materials

Component	Material	Quantity	Colour	Dimensions (mm)	Cost (SGD)
1. Container Lid	Polypropylene	1	White	Ø132 x 25	\$22.12
2. Container body	Polypropylene	1	White	Ø125 x 132	
3. User interface	Acrylic	1	White	92 x 72 x 30, 12.5 edge fillet	\$0.50
4. Housing body	PLA	1	Black	92 x 72 x 50, 12.5 edge fillet	\$3.00
5. Red LED	Multiple	1	Red	Ø5	\$0.25
6. Green LED	Multiple	1	Green	Ø5	\$0.25
7. Rocker switch	Multiple	1	Black	10.5 x 15 x 6.7	\$0.99
8. Diode	Multiple	1	-	-	\$0.19
9. Pump	Multiple	1	White	Outlet outside Ø7.5 Outlet inside Ø5	\$3.00
10. Wires	Copper	11	Red/Black	-	\$0.05
11. Arduino Nano	Multiple	1	Blue	45 x 18	\$31.07
12. USB Cable	Multiple	1	Blue	200 (length)	\$6.00
13. Plastic tubing	Nylon	5	Clear	Ø10 outer Ø8 inner Variable length	\$1.30
14. Plastic tube connectors	Polypropylene	3	White	Ø7	\$0.28
15. Grommet	Rubber	1	Black	Ø12 outer Ø9 hole	\$2.49
16. Screw inserts	Stainless steel	4	Gray	Ø8.75 x 20 + Ø6.1 x 140	\$0.05
17. M4 by 12mm countersunk hexagonal head screw	Stainless steel	4	Gray	12mm head	\$0.05 Total cost: \$71.49

Design Proposal

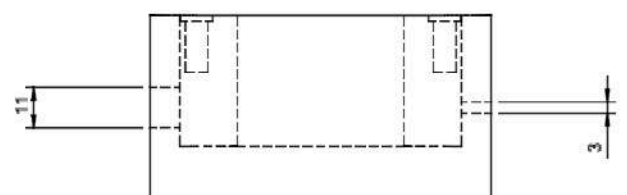
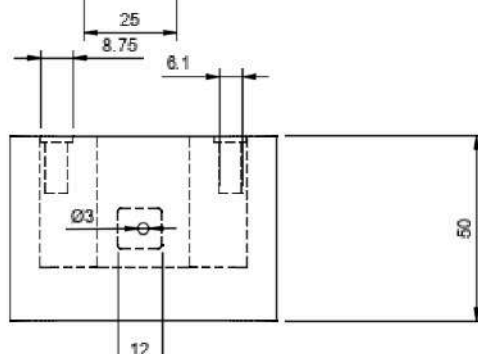
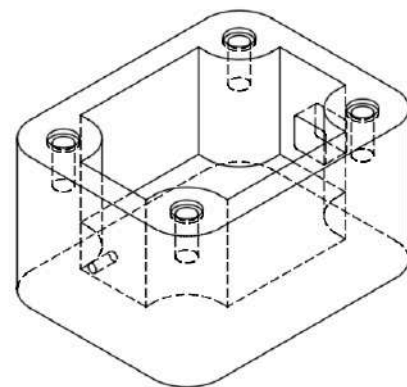
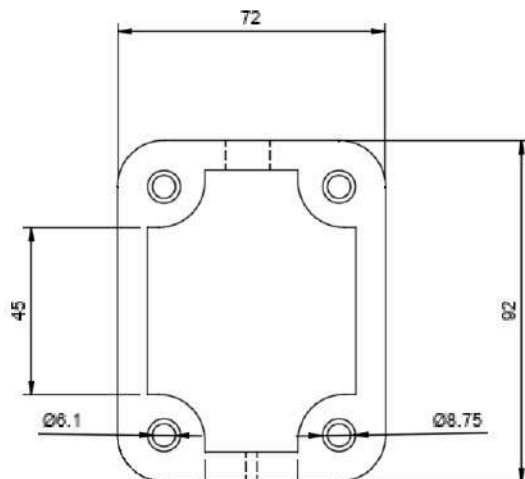


Component name: Full product

Materials: -

Units: mm

Quantity: 1

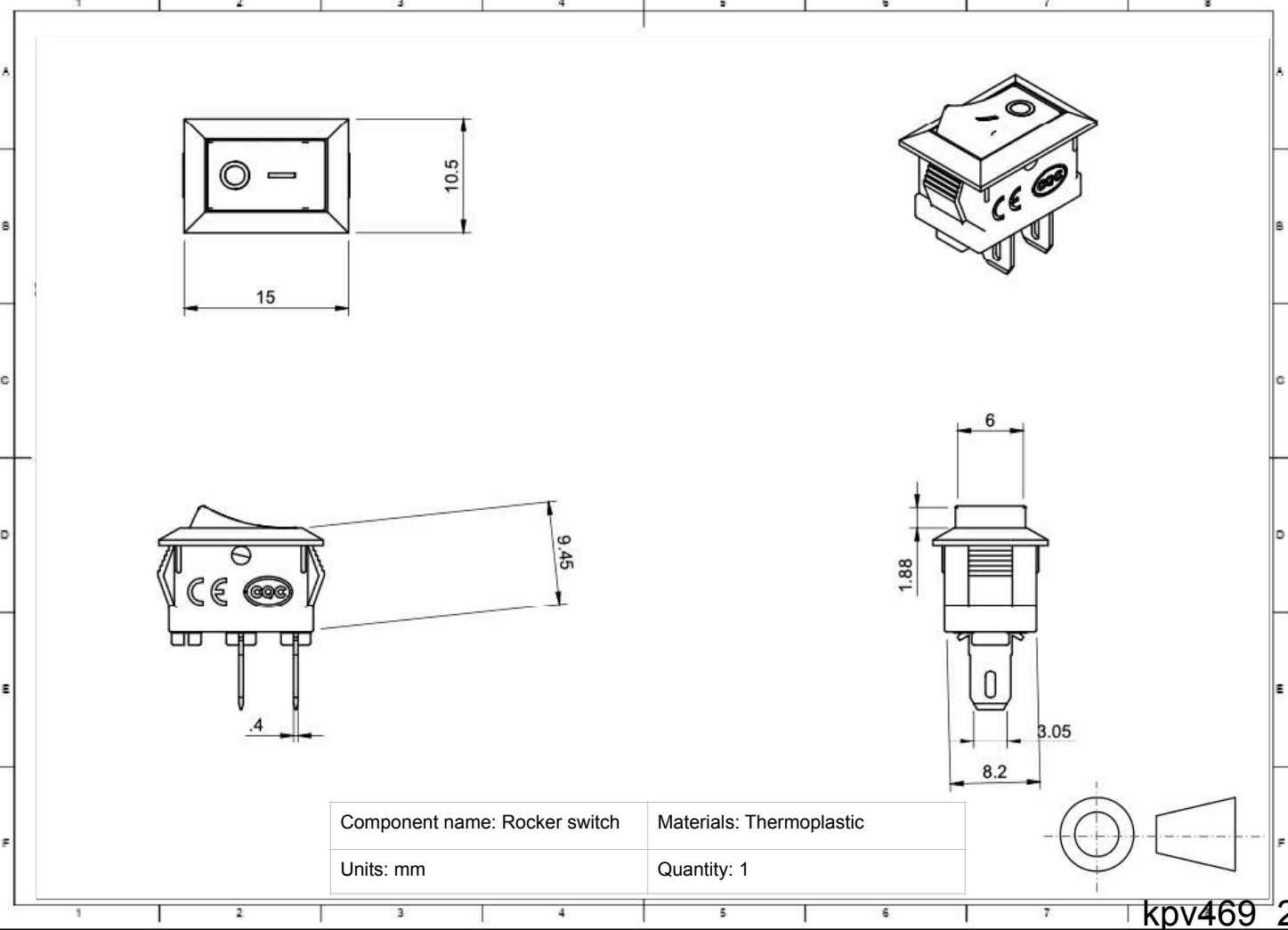
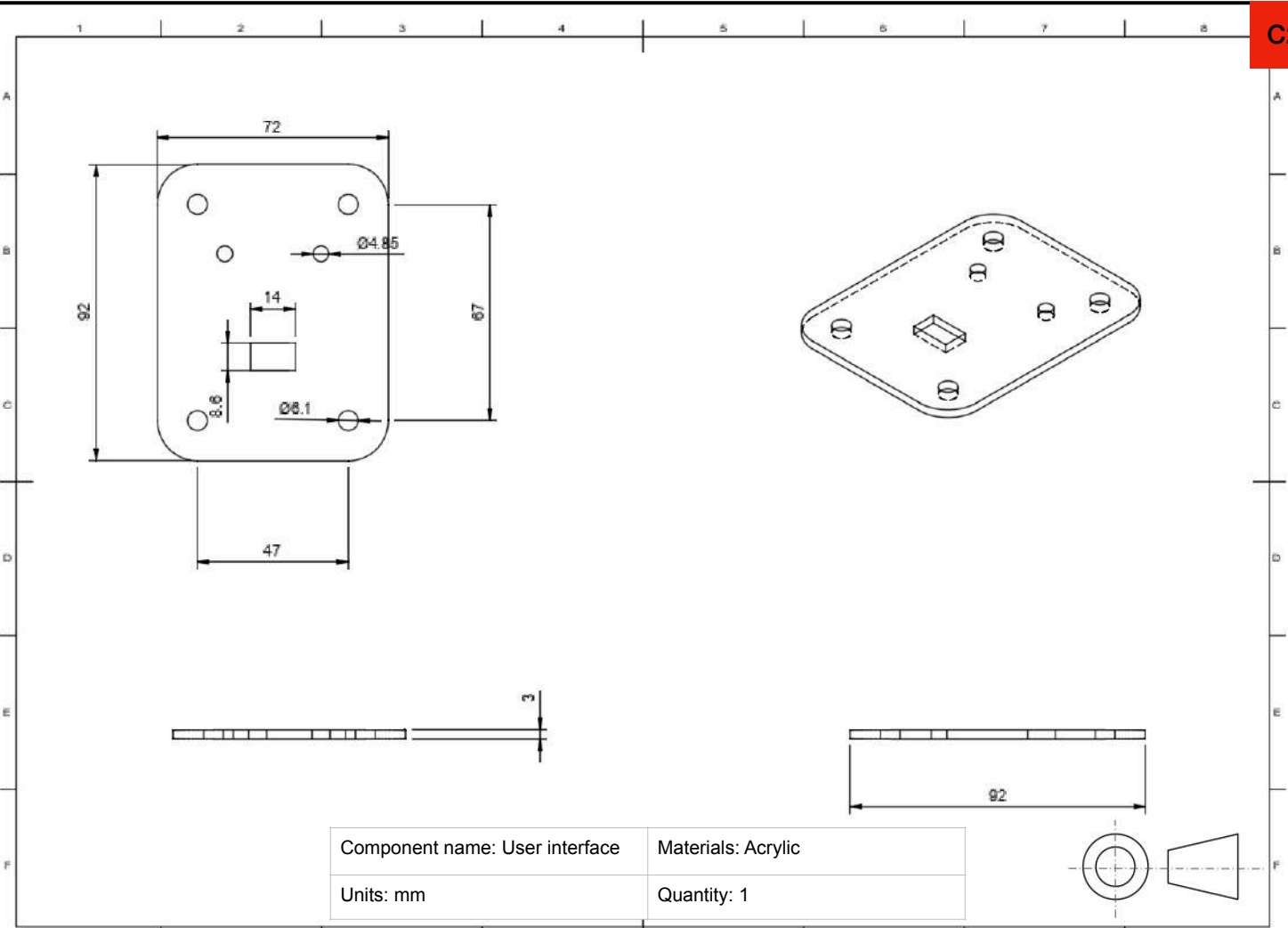


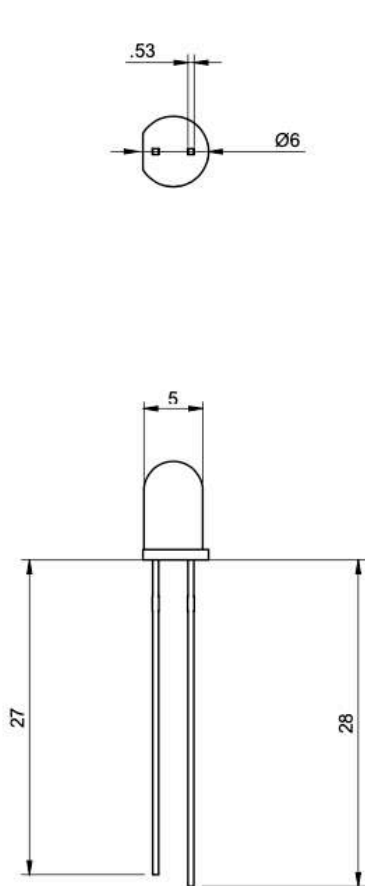
Component name: Housing

Materials: PLA

Units: mm

Quantity: 1



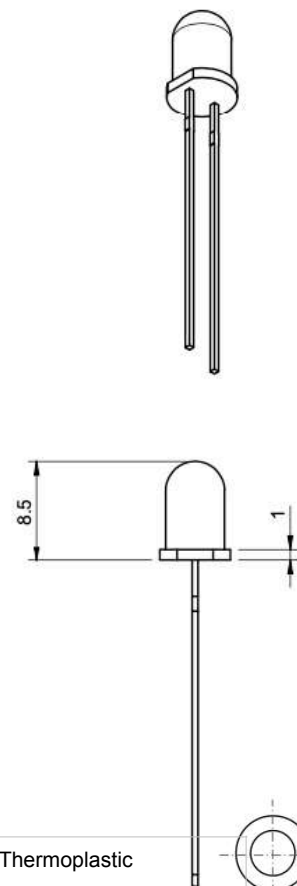


Component name: Green/red LED

Materials: Thermoplastic

Units: mm

Quantity: 2 (1 of each colour)

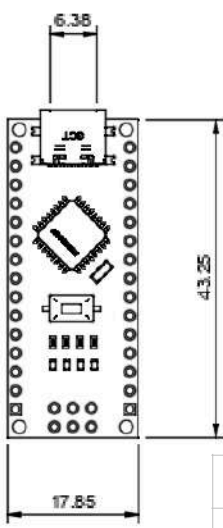
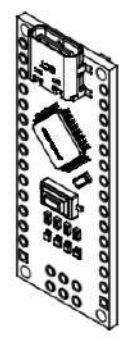
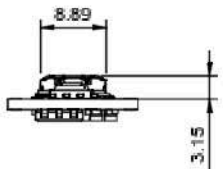


Component name: Grommet

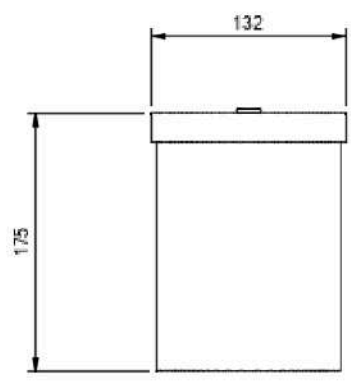
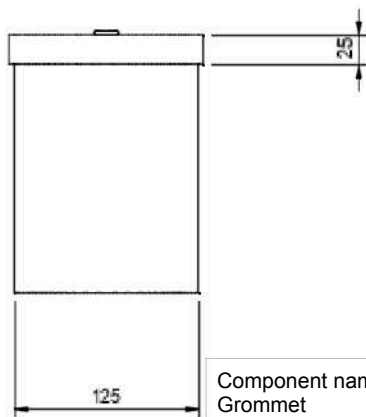
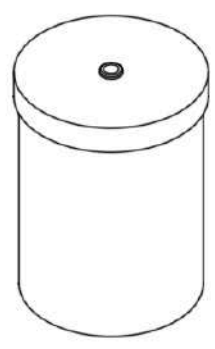
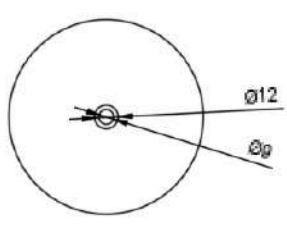
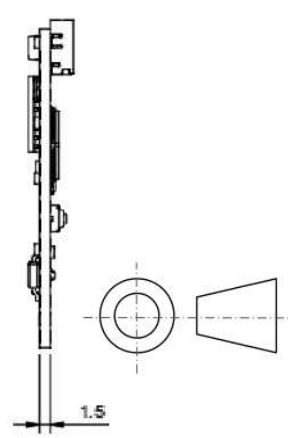
Materials: Rubber

Units: mm

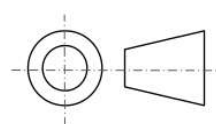
Quantity: 1

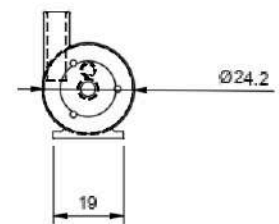
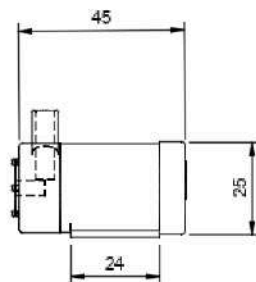
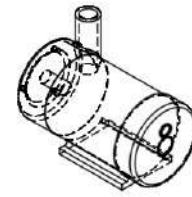
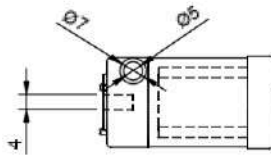


Component name: Arduino Nano	Materials: Multiple
Units: mm	Quantity: 1

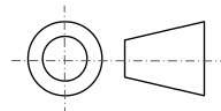


Component name: Water Tank + Grommet	Materials: Polypropylene
Units: mm	Quantity: 1





Component name: Water pump	Materials: Multiple
Units: mm	Quantity: 1



Plan of Manufacture

Process	Equipment	Scheduling	Quality Control	Risk Assessment
1. Creating CAD for housing Create design on CAD software	- Computer - CAD software e.g. Fusion 360	30 mins	- Ensure components are to scale - Be aware of units	- No risk
2. 3D printing Export file from CAD software to 3D printing software. Adjust appropriate settings, connect to 3D printer and print	- Computer - 3D printing software e.g. flashprint - 3D printer - PLA	5 mins + 4 hours	- Ensure the file is exported correctly - Ensure all the 3D printing settings are optimised - Do not move the 3D printed product until fully printed	- Make sure PLA is correctly loaded in 3D printer
3. Adding inserts Use clamps to add inserts into the appropriate holes as it is a tight fit	- 3D product - 4 inserts - Clamp	5 mins	- Ensure the inserts are perpendicular to the product for a perfect fit	- Be careful not to drop the clamp - Be careful not to clamp fingers
4. Creating user interface Create design on software	- Computer - Software e.g. Fusion 360, Adobe illustrator	20 mins	- Ensure drawing is to scale - Be aware of units - Make sure drawing matches housing - Components holes should be made to be friction fitted	- No risk

Process	Equipment	Scheduling	Quality Control	Risk Assessment
5. Laser cutting Export file to laser cutter	- Computer - Software - 3D printer	10 mins	- Ensure the file is exported correctly - Ensure the laser cutting is positioned correctly and set to the correct settings	- Ensure the machine is closed when laser cutting - Don't peel off the protection until after the next step
6. Colour in text Use a marker to colour over the text and peel off the protection after letting the marker dry	- User interface - Marker	10 mins	- Ensure the colour of the marker is right - Make sure all the text is filling	- Wait enough time for the marker to dry before peeling off the protection
7. Creating a countersink Use a 3.5mm drill bit and a hand drill to create a countersink for each of the 4 holes in the user interface	- Hand drill - 3mm drill bit - User interface	10 mins	- Ensure the drill bit is the correct size - Drill down perpendicular to the user interface - Test by placing screw and doing a touch test to feel whether or not the countersink is far enough	- Make sure the user interface is in place when drilling - Keep hands away from drill bit when drilling
8. Creating and importing the code Connect the Arduino nano to the computer using a cable and type out the code for a timed pump using the Arduino IDE software. This can change when testing later	- Computer - Cable - Arduino - Arduino IDE software	10 mins	- Ensure the code is correct - Make sure the Arduino Nano is connected properly and the code is uploaded correctly	- Make sure the Arduino is safe and not at risk of damage
9. Creating and soldering the circuit Create a circuit of the diagram above except for the pump and solder components together. Add heat shrink where metal is exposed and to the unsoldered wire (going to pump) to prevent short circuiting. Use a hot glue gun for areas that can't be soldered e.g. LED and male - female wire.	- Wires (including male - female wires) - Arduino Nano - Red LED - Green LED - Rocker switch - Diode - Soldering iron - Solder - Soldering station - Heat shrink - Glue gun	30 mins	- Ensure the circuit is correct - Copper on circuit shouldn't be visible after soldering - Ensure each component is in correct place - Use a correct sized rubber for the heat shrink - Ensure the glue gun is loaded	- Avoid burning yourself - Avoid inhaling smoke released from solder - Wear goggles - Consider using the vice to clamp the edge of the circuit board for more efficiency and precision - Avoid burning yourself when heating the rubber or when glueing
10. Test the circuit Temporarily connect the pump and test if the circuit works	- Finished circuit - Pump	1 min	- Ensure the pump is connected - Troubleshoot if the circuit doesn't work	- Be gentle with the circuit to not break it

Process	Equipment	Scheduling	Quality Control	Risk Assessment
11. Combining housing with circuit Put the circuit in the housing and have the unsoldered wires (going to pump) come out through the 3mm hole and have the power supply connected to the Arduino Nano come in from the other hole. Push components (LEDs and rocker switch) through the user interface and screw the interface on top	- Finished circuit - Finished user interface - Finished housing	10 mins	- Ensure all components are in place	- No risk
12. Adding grommet Marker out the center of the lid with a marker. Then, drill a hole using a hand drill and a 13mm drill bit, then insert the grommet	- Container lid - Grommet - Hand drill - 13mm drill bit - Marker	5 mins	- Ensure the drill bit is the correct size - Drill down perpendicular to the container lid	- Make sure container lid is in place when drilling - Keep hands away from drill bit when drilling
13. Connecting the pump Push the pipe and wire through the grommet (from out to in) and connect the pump to the wire using solder and the pipe.	- Container lid - Pump - Finished circuit - Soldering iron - Solder	10 mins	- Ensure the wire and pipe are going from outside in - Make sure the tube is tightly secured and connected to the pump	- Avoid burning yourself - Avoid inhaling smoke released from solder - Wear goggles - Consider using the vice to clamp the edge of the pipe for more efficiency and precision
14. Connect the tubes to connectors and additional tubes Connect tubes with connectors such that there are 3 water outputs. Base the lengths of measurements according to the plant locations	- Housing+tube - Additional tubes - Tube connectors - Ruler	3 mins	- Make sure the tube lengths are neither too long nor too short	- No risk
15. Add water and test everything Pour water into the container, connect cable to power source. Measure the amount of water released and adjust the code accordingly to match timings	- Finished product - Water - Power source - Computer - Arduino IDE software	2 mins	- Ensure everything is connected correctly - Troubleshoot if product doesn't work	- No risk
16. Attaching pump to the bottom of water tank If everything else is complete and working, use a hot glue gun to glue the pump to the bottom of the housing (inside)	- Finished product - Pump - Hot glue gun	3 mins	- Ensure everything works before proceeding - Ensure the pump is correctly stood up (flat side at the bottom)	- Avoid burning yourself when glueing

Non destructive tests	Testing method	Justification
User trial (1.1, 1.2, 1.3, 3.4, 7.2) Observability: ease of arranging pipes and powering on	- Provide a scenario: pipes need to be diverted to the plants and power needs to be turned on - Constraint: complete under 1 minute	- Pipe management should be straightforward and require minimal time - Powering the product should be the easiest step
User trial (1.1, 1.2, 1.3, 3.4, 7.2) Observation: adjustability of time on Arduino code	- Provide a scenario: a plant needs to be watered every 6 hours - Constraint: complete under 5 minutes	- User should be able to adjust timing according to plant needs - Code should have high learnability as it is an important aspect of the product
User trial (1.5) Observation: refilling water tank	- Provide a scenario: water tank needs to be refilled from empty - Constraint: complete under 1 minute	- Pipe management should be straightforward and require minimal time - Powering the product should be the easiest step
Performance test (1.4) Measure whether water is evenly distributed	Power the product and use water containers to collect water at the end of each pipe in cups	- Water distribution should be even - An uneven water distribution could lead to over and under watering
Performance test (2.3) Measure the weight of the product and water tank with and without water	- Weigh the water tank on a scale with and without water - Weigh the housing	The product should be light enough to be portable when needed and stable when not moving
Performance test (2.1, 2.2, 5.1, 5.2) Measure the dimensions of the product	Measure the dimensions with a ruler	The product should be compact enough to not block sunlight nor hinder the plants growth
Performance test (1.6) Measure how long the water lasts	- Allow the pump to run until the tank is empty - Measure the time it takes for the 2L to empty	The user should understand approximately how long 2L will last for each code/delay
Destructive tests	Testing method	Justification
Performance tests (4.1) Waterproof test	Spraying the product with water whilst in operation and checking if any water leaks in	- User error could potentially lead to spillages - Hence, it is important to have the product be waterproof in case of spillages
Performance tests (4.3) Strength and toughness test	Apply compressive forces to both the tank and the body	- The product should be able to withstand basic forces such as moving the product - User error could potentially lead to the product being dropped - Hence, it is important for the product to withstand basic forces

Evaluation Against Specifications

D2

Note: client has moved home however, plants are placed at a similar location in the new apartment with similar dimensions. Plants have also changed to 3 basil plants

Non destructive tests - User trial

Client attempting to operate the product and arrange the pipes



Observations:

- Client finds the power switch easily
- Client recognises the product is on due to LED lights
- Tubes are hard to arrange as they revert and often fold, hence client has to use clips

Client attempting to change the Arduino code



Observations:

- After the first attempt, the client spent less than 2 minutes changing the code
- Client took a bit of time changing the code on the first try, (>5 minutes setting up app and reading code)

Client attempting to refill the water tank

First trial



Observations:

- Client easily comprehends how to unscrew lid
- Water tank wasn't heavy for client even after filling
- Troublesome to remove lid due to having to slide pipes and wires
- Refilling the tank is tough since there was no tap nearby and the client had to bring the whole product to refill

Second trial



Observations:

- Product can be kept at its location
- Tubing won't have to be removed and setup can remain as it is
- Multiple trips will have to be made to fill tank

Non destructive tests - Performance tests

Weight of the product at different states



Results:

Full water tank: 2196.8 g

Empty water tank: 215.7 g

Housing with component: 126.6 g

Significance of the results:

These results show that the housing and its components are extremely lightweight. The tank when full is slightly heavier but still portable

Distribution of water



Results:

The water was slightly uneven as shown by the adjacent photo after 30 seconds of pumping. Most of the water ended up at the bottom plant due to the different elevations

Significance of the results:

This means that the water is not distributed evenly and can result in overwatering and underwatering

Dimensions of the product



Results:

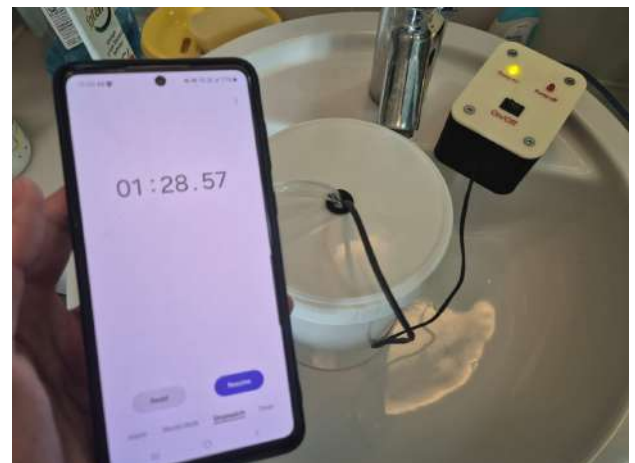
Housing: 9.2 x 7.2 x 5 cm

Water tank body + lid: Ø125 x 132 + Ø132 x 25

Significance of the results:

This demonstrates that the product doesn't take up much space. It should also be noted that the water tank and housing can be moved freely to a certain extent

Duration the product lasts



Results:

It took approximately 1 min 30 to fully pump out 2L. Water is pumped out at a rate of 80L/h = around 1.3L per minute

Significance of the results:

This will determine how long the water in the tank can last depending on how the user changes the code

Destructive tests - Performance tests

Waterproof test



Results:
No water enters the housing from the top

Significance of the results:
This is crucial as accidents may lead to spillages.
A sealed housing means the components inside don't get destroyed

Strength and toughness test



Results:
No failure under pressure by human grip (squeezing and stretching)

Significance of the results:
This ensures that the product is strong enough and will not fail under regular forces

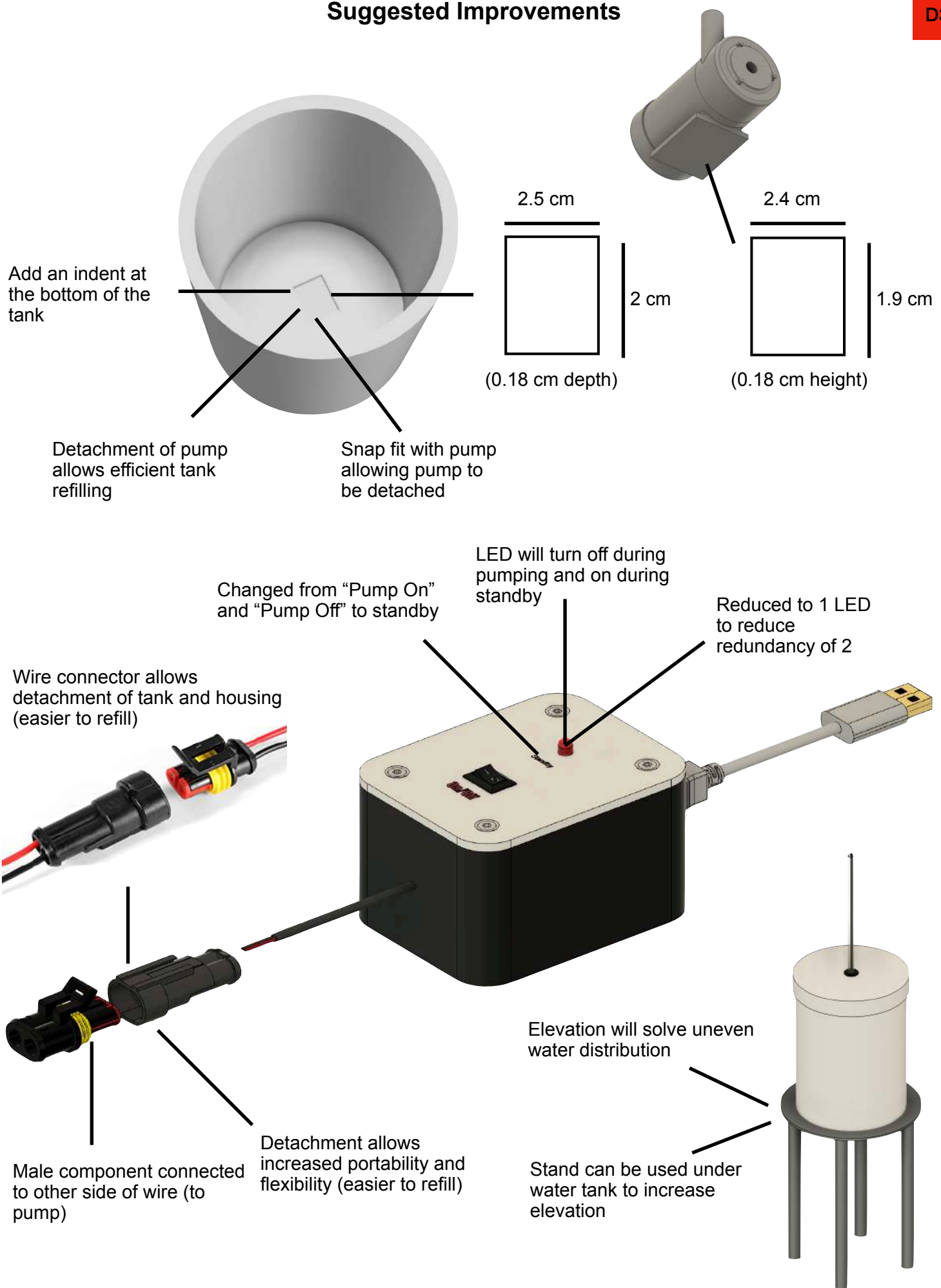
Summary of testing in D2

Green: specification met, Orange: specification somewhat met, Red: specification not met

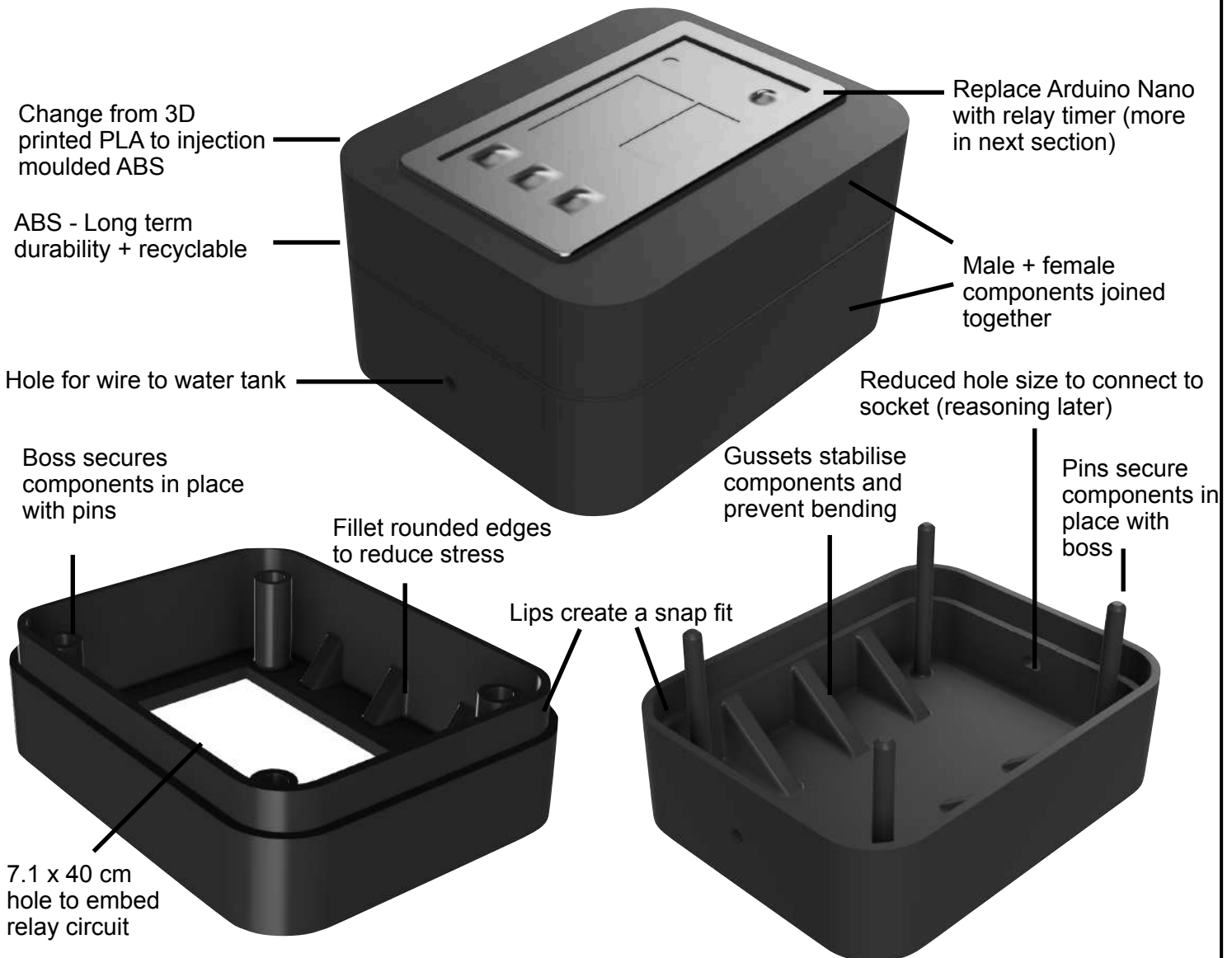
- 1.1: Water is pumped automatically when the client turns on the product
- 1.3: Product can be adjusted by adjusting the Arduino code
- 2.2: Product doesn't block sunlight
- 2.3: The total weight of the product was not over 10kg
- 3:4: User interface is simple to understand (simple on/off switch) and didn't cause the client any trouble
- 5.1: Product doesn't occupy a space more than 55 cm
- 1.2: Water has the ability to water plants on 3 different levels but water is not distributed evenly
- 1.6: Water will be able to last 3 weeks depending on the user setting
- 2:1: Product slightly hinders plant growth due to the tubes
- 5.2: Water can provide water to the specified area, however, with uneven distribution
- 7.2: Product was easy to understand and use for the client besides arranging the tubes
- 1.4: Product doesn't distribute water evenly due to the different elevations of the plant
- 1.5: Water is fairly challenging to refill, even after testing 2 different methods

Suggested Improvements

D3

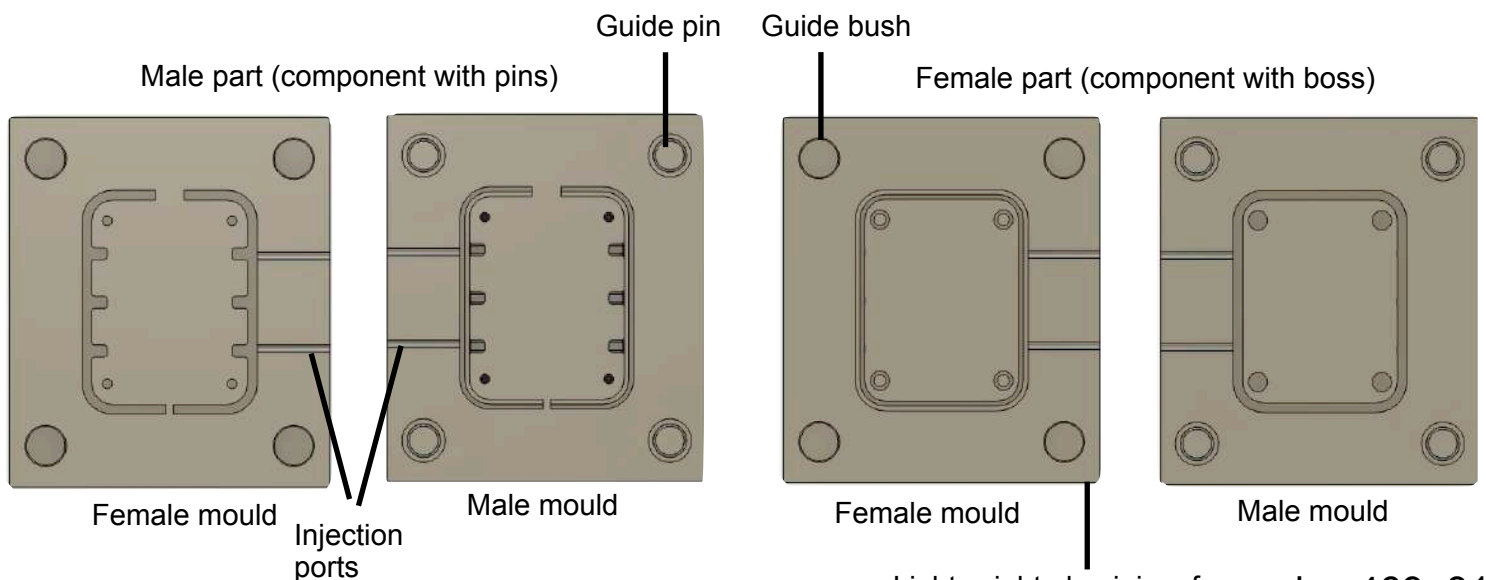


Changes to housing to allow for commercial production

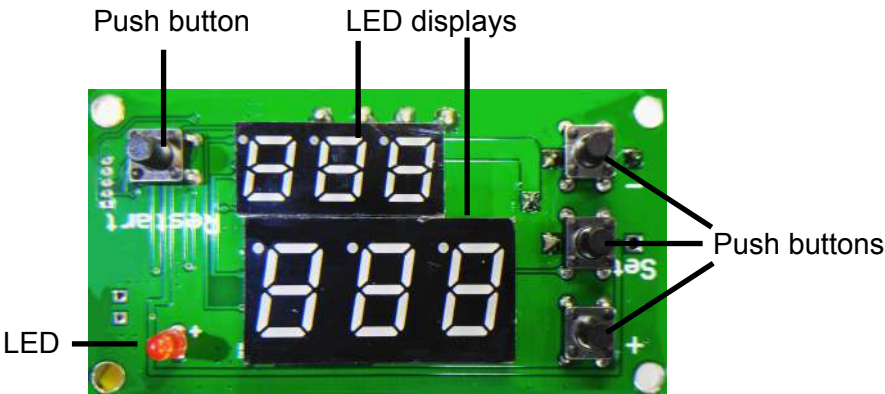
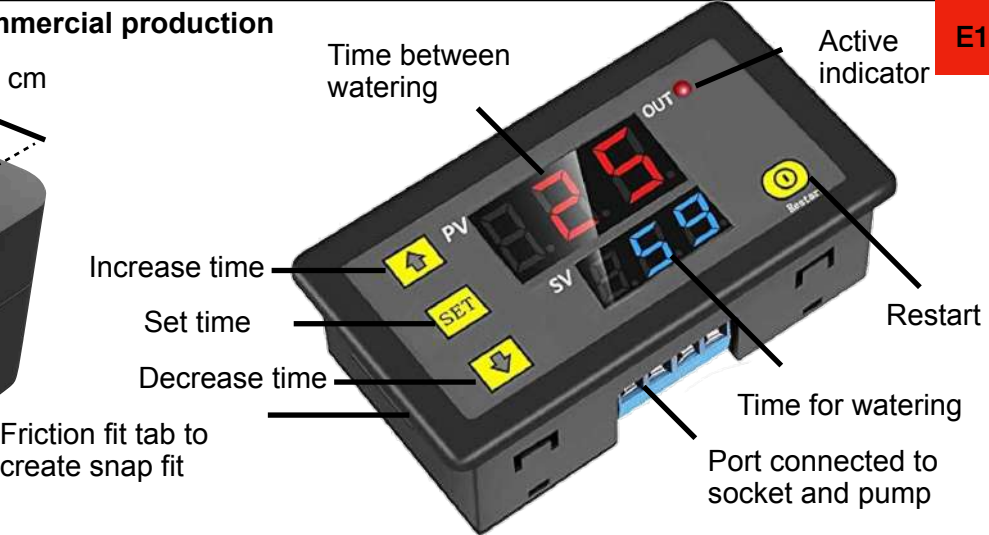
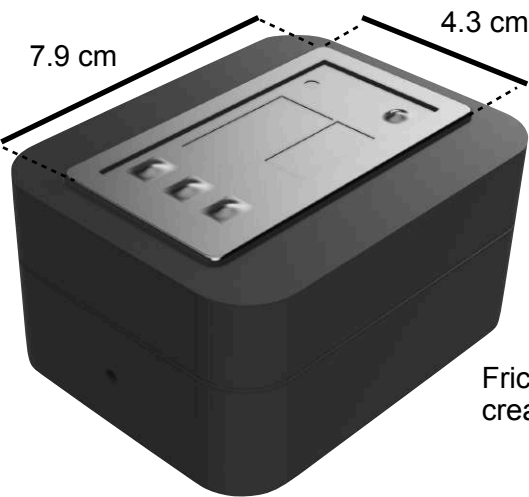


Commercially viable production process

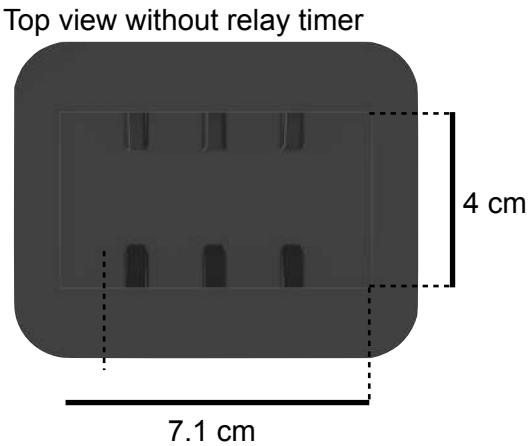
The process of injection moulding requires a mould which was created on a CAD, Fusion 360. The use of CAD and the mould was used to carry out an analysis of mould flow in section F2. This is important to understand how the adjustments will allow my design to be injection moulded. Note that a 3° angle has also been implemented in the mould to facilitate the removal of the product from the mould, reducing the likelihood of damage.



Changes to circuit to allow for commercial production

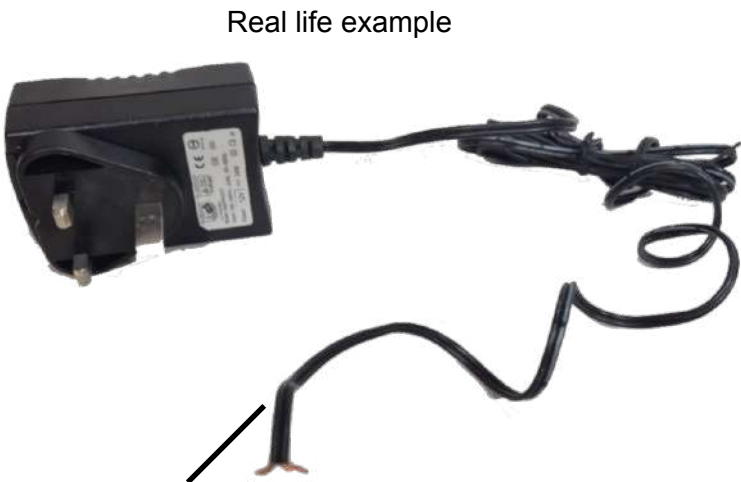
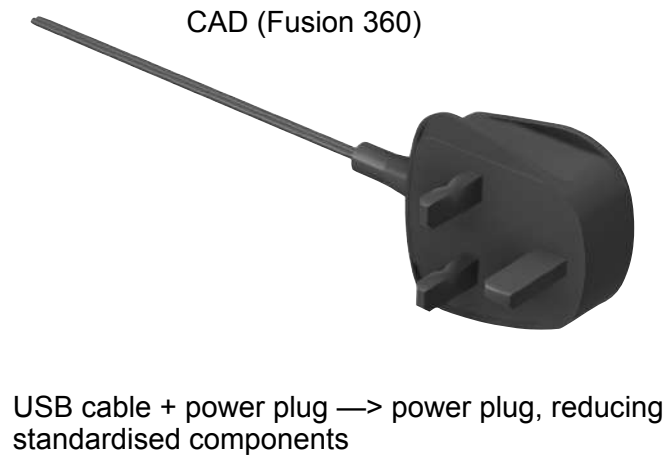


Working Voltage	DC 12V
Timing Range	0-999s, 0-999m, 0-999h;
Function	Timer, Delay, Cycle;
Maximum Current	10A
Maximum Load	15000W
Mounting Size	71*40*24 mm
2.79*1.57*0.94 Inch (L*W*H)	

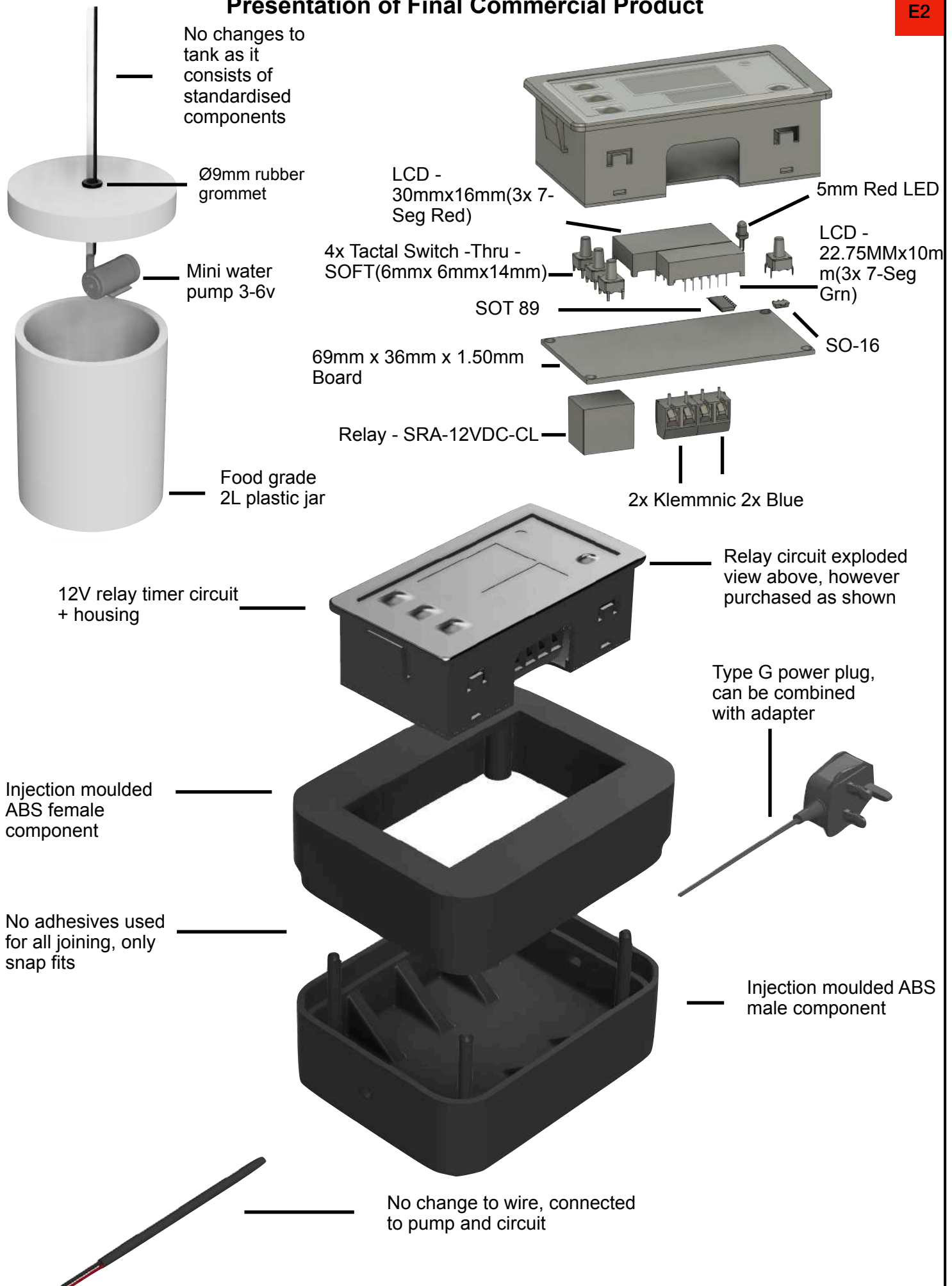


Changes to the circuit will optimise the use of standard components by replacing the Arduino Nano + circuit components to a singular circuit with simple snap fit installation.

Changing the circuit will require a power plug change. The circuit requires 12 volts to operate, making a power outlet more applicable instead of a battery. In addition, the housing should be as compact as possible to reduce costs when injection moulding. A type G power plug shown, however an adapter can be used to match the target user's needs.



Presentation of Final Commercial Product



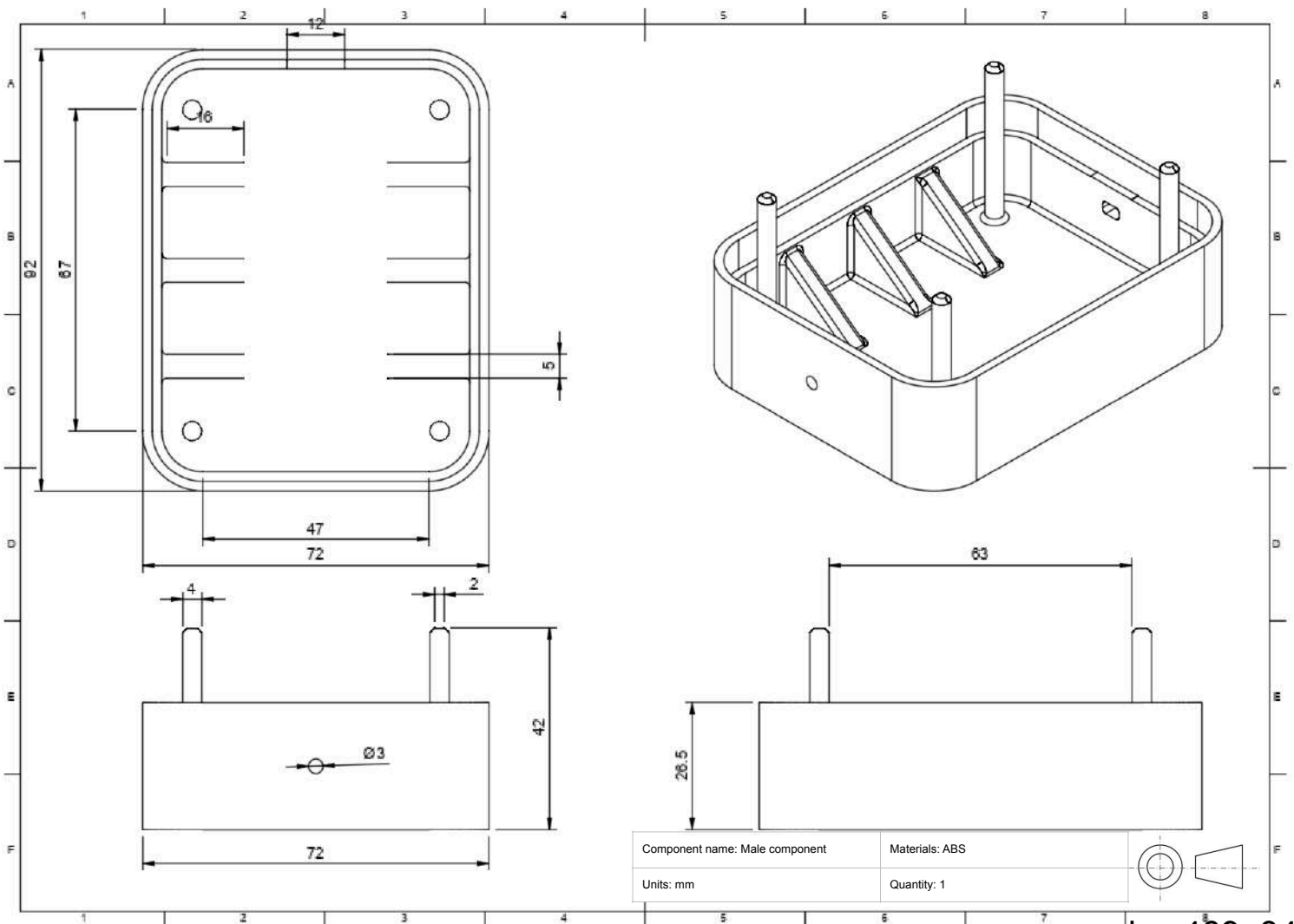
Third Party Product Manufacture Details

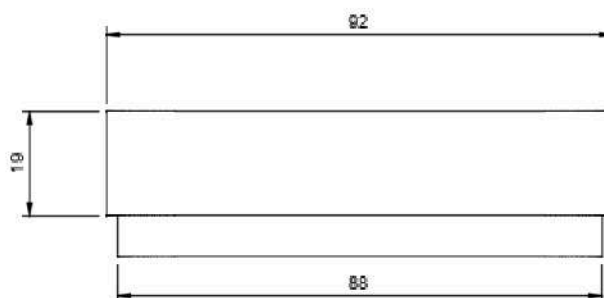
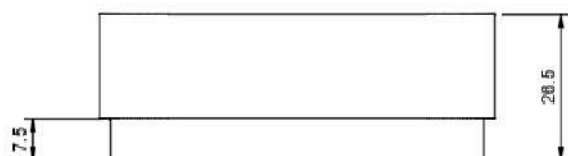
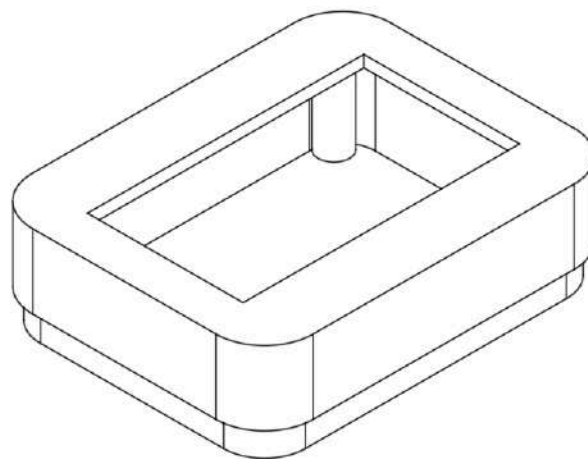
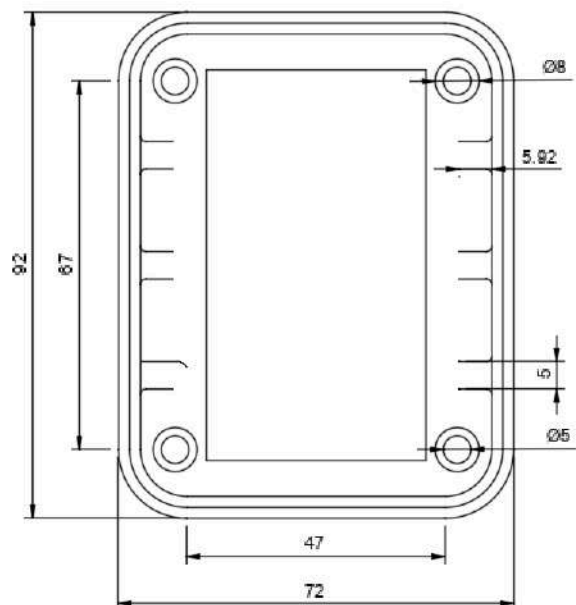
E3

Part	Description	Quantity	Dimensions (mm)	Unit cost (SGD)
Injection moulded ABS female component	ABS	1	92 x 72 x 30, 12.5 edge fillet	\$0.67
Injection moulded ABS male component	ABS	1	92 x 72 x 50, 12.5 edge fillet	\$0.67
Type G power plug	Multiple	1	-	\$1.50
12v Cycle Delay Timer relay	PCB + Plastic	1	Housing: 79 x 43 Mounting size: 71 x 40 PCB: 69 x 36	\$8.72
Food grade 2L plastic jar	Polypropylene	1	Body: Ø125 x 132 Lid: Ø132 x 25	\$\$22.12
Mini water pump 3-6v	Multiple	1	Outlet outside Ø7.5 Outlet inside Ø5	\$3.00
Ø9mm rubber grommet	Rubber	1	Ø12 outer Ø9 hole	\$2.49
Ø10mm Plastic tubing	Nylon	1	Ø10 outer Ø8 inner Variable length	\$0.28
Ø8mm Plastic tube connectors	Polypropylene	3	Ø7 outer	\$1.30

Note: cost shown is for one unit, costs will be significantly when ordered in batch

Total cost: \$40.45



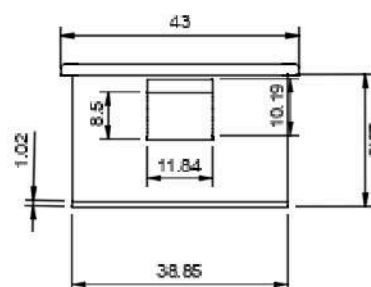
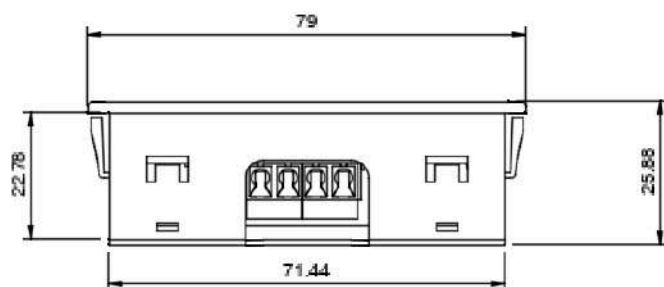
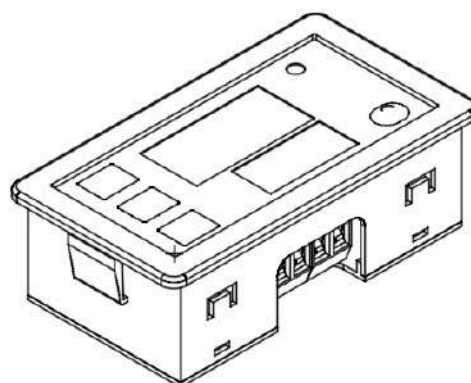
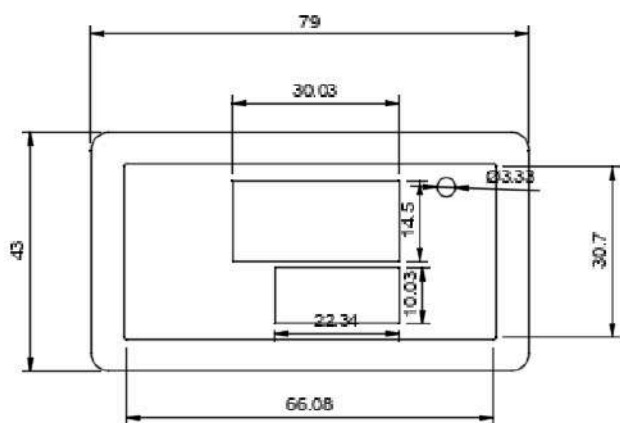
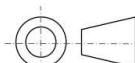


Component name: Make component

Materials: ABS

Units: mm

Quantity: 1



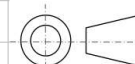
Note: water tank isometric have not been shown in E3 as it is identical to C2

Component name: 12v Cycle Delay Timer relay

Materials: Multiple

Units: mm

Quantity: 1



Materials

ABS (Acrylonitrile Butadiene Styrene); components - male and female housing

Choosing ABS for injection moulding in commercially produced products offers a multitude of advantages. Firstly, ABS can be easily moulded and has a low shrinkage rate during cooling which ensures precise replication and standardisation of the product when batch produced. In addition, it is a lightweight material, meaning it can be transported with lower costs. Finally, ABS is a thermoplastic that can be reused or recycled when melted.



ABS pellets in multiple colours

Components

12v cycle delay timer relay

Circuit used for adjustability (range of timings) and used as a standardised component.

Key components within circuit

5mm Red LED - indicator for when circuit is closed

2x LCD - display feedback for user

4x tactical switch - allows user to adjust time circuit is open and closed

Relay - open and closes the circuit



Power plug

Required to power 12v cycle delay timer relay. Connects circuit to power outlet and eliminates the need of a battery. Also used to power on and off the circuit.



Food grade 2L plastic jar

Used to store water, comes with an unscrewable lid to allow for refilling. Material - made of polypropylene which has great cost effectiveness, strong chemical resistance, and is a thermoplastic which can be recycled



Mini water pump 3-6v

Required to pump water to the plants. Active when the circuit is complete, which can be adjusted with 12v cycle delay timer relay



Ø9mm rubber grommet

Required to keep water tubing and wiring in place on the plastic jar. Also needed to minimised water evaporation



Ø10mm plastic tubing

Needed to transport water from pump to plants as well as connecting to plastic tubing



Ø8mm plastic tube connectors

Required to split and distribute water to plants by connecting plastic tubing



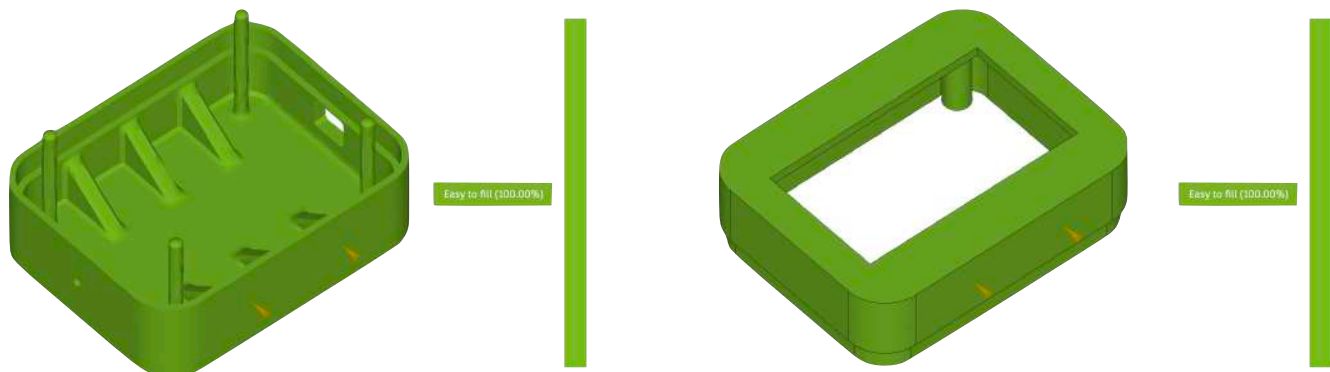
Injection moulding

The male and female parts will be injection moulded as it is an automated fast-industrial process that can produce complex 3D shapes. Injection moulding is also optimal for high volume production methods such as batch production as it requires minimal labour costs and produces little waste, with no finishing required. However, an expensive one time cost mould must first be created.

The adjacent photo represents the conditions that will be used for the simulation. The 2 injection locations are positioned to match the ones shown in the mould above. The recommended melt temperature and mould temperature for ABS are 210°C to 260°C and 45°C to 80°C respectively (Fictiv). Lustran ABS 1146 by INEOS ABS will be used as the simulation material as it is a popular ABS choice for manufacturing.

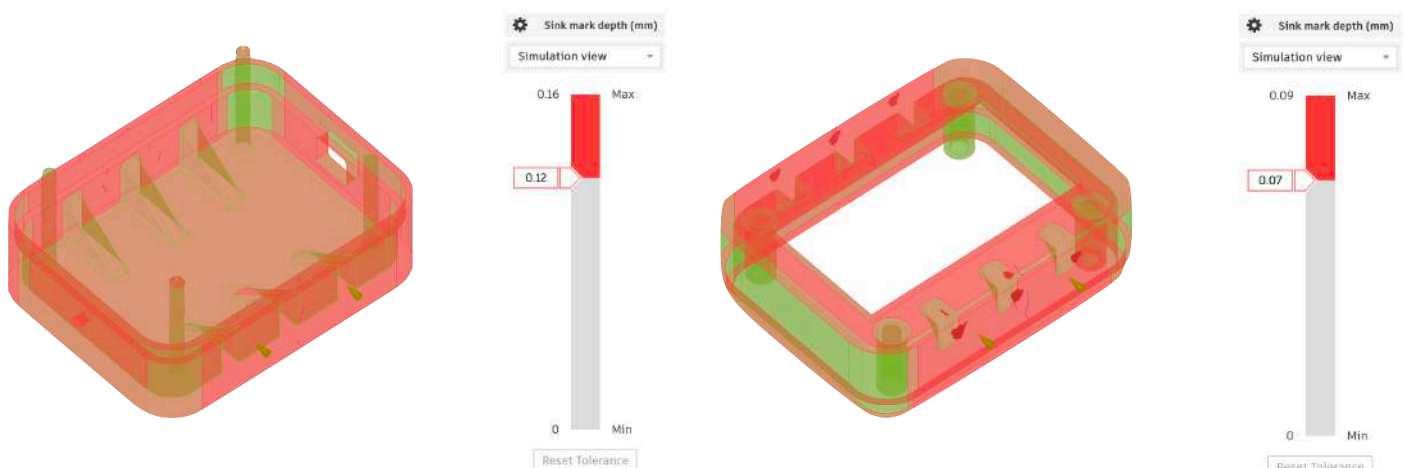
Material	Lustran ABS 1146 : I...	
Injection Locations	2	
Aesthetic Faces	0	
Injection Time	Automatic	
Melt Temperature	260.00 C	
Mold Surface Temperature	80.00 C	

Injection mould fill simulation



Above are the simulation results when injection ports are placed as shown (simulating mould). Since the results display 100% easy to fill, this means that the creation of this component should always be reliable if the heating of plastic is done appropriately and is injected at a constant rate. Injection moulding should yield consistent filling results if the mould remains constant.

Visual defects simulation



The simulation above represents the visual defects for each housing, with each simulation representing the sink marks and weld lines. Note that the tolerance has been set to default by the CAD simulation. Increasing the tolerance decreases the number of sink marks whilst weld lines remain constant. Hence it is evident that the model can be improved to reduce sink marks and weld lines by increasing the tolerance. This can be done by moving the injection ports to thicker locations, reducing the thickness of the walls or creating a smoother surface finishing. Melt and mould surface temperature are already maximised for ABS, hence they cannot be increased further to increase tolerance. Injection moulding is the ideal production process as variables can be adjusted to minimise defects.

Market size

It was discovered in section A that approximately 66% of Americans own at least one indoor plant whilst around 73% travel at least once a year, meaning a potential market size of approximately 159 million customers solely in America.

Batch production vs mass production

Batch production and mass production are the main methods of production. I will be comparing the pros and cons of both methods to justify my method of production.

Batch production (product produced in batches)	Mass Production (continuous production)
+ User acceptance can be measured based on the time it takes to sell a batch which can be used to adjust the quantity of the next batch + Easier to maintain high quality due to smaller batches + Lower initial costs + Reduced risk of excess inventory and obsolescence	+ Higher production rate which often leads to quicker time to the market + Economies of scale can be used + Shorter lead times due to streamlined process + Efficient use of resources due to high volume
- Slower production rate compared to mass production - Longer lead times for manufacturing and delivery - Higher per unit costs (no economies of scale)	- High risk for new products, if not accepted quickly, it will require excess inventory - Higher initial costs due to investment in capital - Limited room for experimentation due to high volume

Comparing both methods of production, batch production is clearly advantageous for a new product entering the market due to low costs and risks as well as great flexibility for experimentation and batch sizes. Therefore, it would be ideal to start with a small batch such as 100 and monitor its acceptance to adjust the quantity of the following batches.

Just-in-time vs just-in-case

These are two production strategies that prioritise inventory management differently. I will be evaluating both strategies taking into account its compatibility with batch production to justify my strategy choice.

Just-in-time	Just-in-case
+ Minimises inventory levels, reducing storage costs and freeing up capital + Promotes lean production processes with reduced waste and improved efficiency + Enhances responsiveness to market by aligning production with demand	+ Builds inventory to mitigate risks of demand variability and disruptions + Offers flexibility to handle unforeseen events without causing disruptions + Guards against supply chain disruptions and unexpected demand surges
- Vulnerable to disruptions in supply chain or production, which can halt operations - Relies heavily on reliable suppliers and transportation networks for timely deliveries - Limited flexibility in adjusting production schedules or handling sudden demand spikes	- Requires higher levels of inventory, leading to increased carrying costs - May lead to overstocking and tie up capital in inventory that may not be sold quickly

Comparing both strategies, just-in-time appears to be the better option between the two as it minimises storage and costs. As this is a new product entering the market, it will likely take time to be accepted. Hence, having the philosophy of responding to demand rather than storing extra in case of an influx demand is appropriate as the initial demand will likely be low. In addition, it was mentioned that the following batches would be based on how quickly the product sells out, hence, this would be more appropriate for a just-in-time method.

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